

# PV–PWR–4 power board

Rev A2 | 2026–10–05 | Schematics authored by Chinmoy Bhuyan | PV–P100/110: 4 interleaved FSBB phases, gate drive, sensing, ports, aux (D–044)

## 01 PC connector, command buffers, status

Interfaces PC (2x32) to the control board, 3.3 -> 5 V command buffers, RDY / FLT\_N, +5 V supervisor, port status glue, BUS- star points

File: sheets/01\_pc.kicad\_sch

## 02 Phase 1 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HO/A 75, NTC inputs, gate bias (gdrv.bias\_phase)

File: sheets/02\_ph1\_power.kicad\_sch

## 03 Phase 1 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/03\_ph1\_gda.kicad\_sch

## 04 Phase 1 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/04\_ph1\_gdb.kicad\_sch

## 05 Phase 2 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HO/A 75, NTC inputs, gate bias (gdrv.bias\_phase)

File: sheets/05\_ph2\_power.kicad\_sch

## 06 Phase 2 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/06\_ph2\_gda.kicad\_sch

## 07 Phase 2 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/07\_ph2\_gdb.kicad\_sch

## 08 Phase 3 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HO/A 75, NTC inputs, gate bias (gdrv.bias\_phase)

File: sheets/08\_ph3\_power.kicad\_sch

## 09 Phase 3 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/09\_ph3\_gda.kicad\_sch

## 10 Phase 3 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/10\_ph3\_gdb.kicad\_sch

## 11 Phase 4 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HO/A 75, NTC inputs, gate bias (gdrv.bias\_phase)

File: sheets/11\_ph4\_power.kicad\_sch

## 12 Phase 4 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/12\_ph4\_gda.kicad\_sch

## 13 Phase 4 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/13\_ph4\_gdb.kicad\_sch

## 14 Port A power path (pv)

Varistor Y + monitor

File: sheets/14\_leanA\_power.kicad\_sch

## 15 Port A sensing and interlocks

References, BUS– shunt amplifier + hold–off / OC windows, dividers, dV amplifier, polarity and dV interlocks

File: sheets/15\_leanA\_ctrl.kicad\_sch

## 16 Port A coil drives

Contactors: interlock logic, fail–safe high side, economiser; passive bank bleeder

File: sheets/16\_leanA\_drive.kicad\_sch

## 16 Port B power path (battery)

CAUTION: not to be used for design\_check() asserts every number on each build (outputs of PWR–4 design check.txt), drafted @ few days from now, at build time. Domains: LIVE (BUS– = GND = AGND), GD\_PxSy gate islands (functional: NSI6651, T\_BIAS4), PE (AIN pads, inductor cores, heatsink NTC lugs, Y / MOV3 / MD crossings), SELV (AUX–T1 reinforced winding -> J\_SELV). Coat to PD1. Shared heatsink: Tj 110 C at 45 C inlet (R\_sa target); heatsink OT trip must be <= 90 C. Dead time 211–554 ns. Sensor delay 0.33 us, error 1.8 % at 45 A after calibration. Live 24 V 23.7 W (economiser) <= aux live rating 30.0 W; module aux 82 W. Full fan speed is only guaranteed while the converter is switching.

File: sheets/16\_leanB\_power.kicad\_sch

## 17 Port B sensing and interlocks

References: PWR–4 design check.txt, drafted @ few days from now, at build time. polarity and dV interlocks

File: sheets/17\_leanB\_ctrl.kicad\_sch

## 18 Port B coil drives

Contactors + precharge relay: interlock logic, fail–safe high side, economiser; passive bank bleeder

File: sheets/18\_leanB\_drive.kicad\_sch

## 18 Terminals, EMI filter, insulation test

DC terminals, nanocrystalline CM ring per port, X / Y capacitors, board entries (press–fit), PE links, heatsink, insulation monitor (port.lean\_imd)

File: sheets/18\_terminals\_imd.kicad\_sch

## 19 Live 5 V / 3.3 V, SELV connector

+24V (aux live) -> +5V and +5V\_GD (LMR38020), +3V3 (TPS62130 from +5V\_GD ramp–up and discharge), +5V\_GD supervisor, SELV connector to the control board

File: sheets/19\_live5v\_selv.kicad\_sch

## 20 AUX75 input taps, OR, film

One tap per port (+ pole, terminal side), return = BUS–: 2 x 20R R pulse–rated ahead of the 1 A / 1000 VDC fuse, BYG10Y OR; 2 x 3.3 uf / 1300 V film = surge absorber (AUX–HV rev C1 input block)

File: sheets/20\_aux75\_input.kicad\_sch

## 21 AUX75 start–up, brown–in, OV, timer

As AUX–HV rev C1 sheet 02: 3 x BSS126 cascode, TPS3710 brown–in 206–215 V, TLV3202 OV lockout 1107–1162 V and overload timer 39–100 ms

File: sheets/21\_aux75\_start.kicad\_sch

## 22 AUX75 PWM, SIC, clamp, AUX–T1

UCC28C59–Q1 64.9 kHz, IV2Q171R0D7Z, Lp 0.485 mH, Np:Nlive:Nselv 50:9:10; live 24 V regulated by the FB divider (no opto); VDD from the live 24 V through the BC817 follower; foldback detector on the live winding

File: sheets/22\_aux75\_pwm.kicad\_sch

Not bench–validated. Values read from sim/out/\*.json; checks: design\_check()

Sheet 1

Schematics authored by Chinmoy Bhuyan

Live board referenced to BUS–; gate islands functional; SELV winding reinforced; HV–PE basic

Sheets: 1

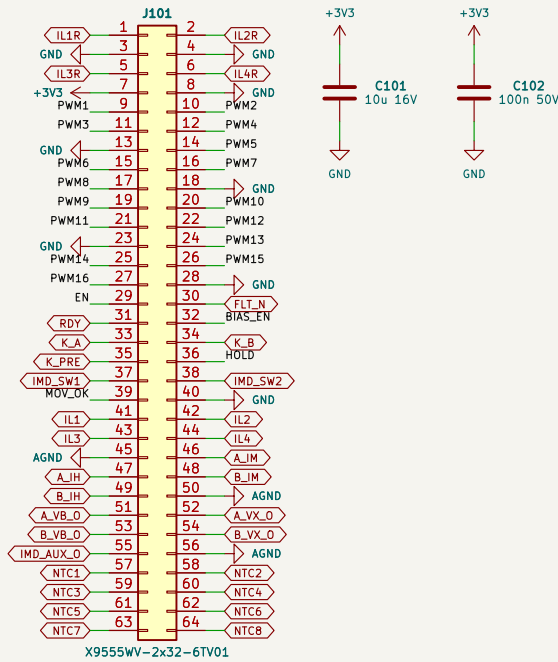
File: F

Title

Size: 6

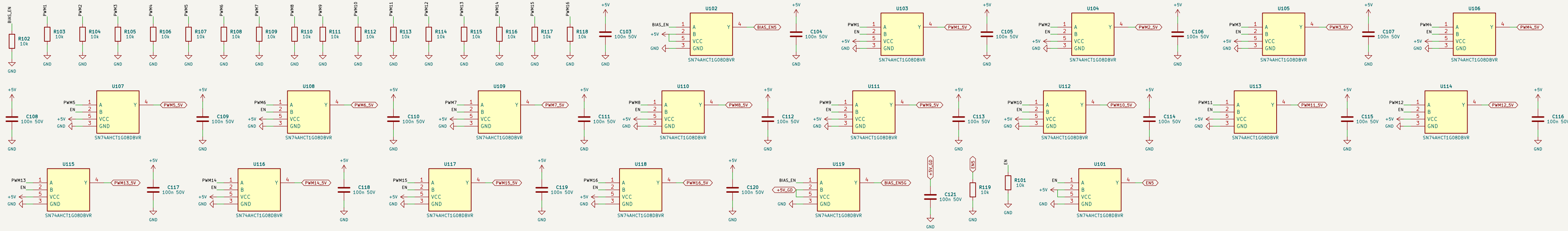
KiCad

PC CONNECTOR (INTERFACES.PC), LIVE



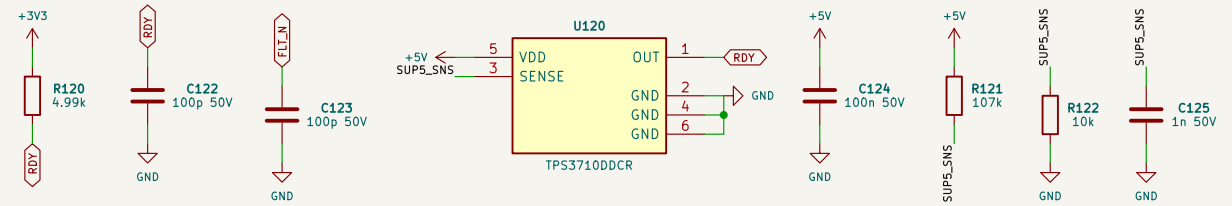
Both boards live: GND = AGND = BUS-, +3V3 goes TO the control board (rev A2: the former +24V / +5V pins carry the sensor references (ILR-IL8)). Commands are 3.3 V logic after the control board's trip latch: every command input is pulled low here (open = off).

COMMAND INPUTS: PULL-DOWNS AND 3.3 -> 5 V BUFFERS (GDRV.INPUT\_BUFFERS)



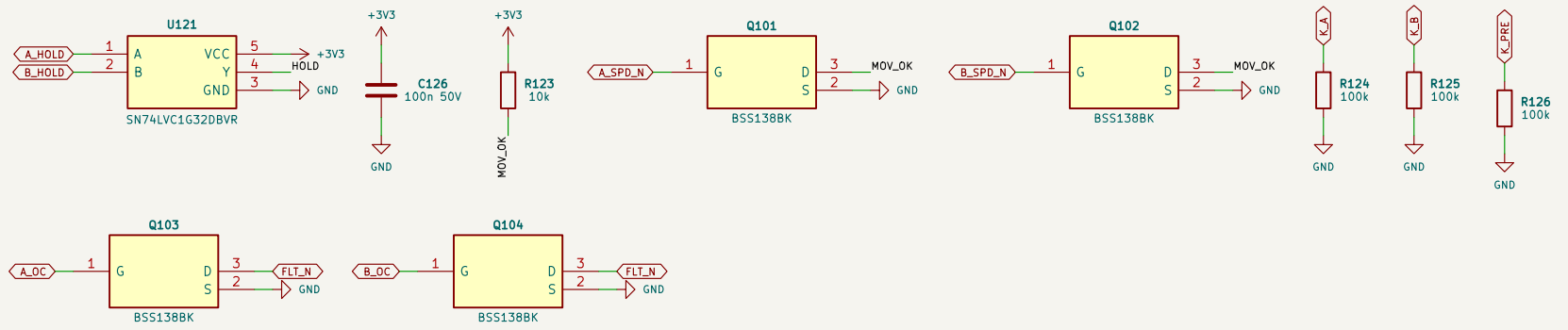
NSI6651 inputs are specified at VCC1 = 5 V only (VIN1 <= 3.3 V, p7) and the SN6505B EN needs 0.7 VCC: one SN74AHCT1G08 (VIN 2.0 V) per line. Each PWM is ANDed with EN here as well, so a low EN removes every command within 8 ns, ahead of the drivers' RST/EN filter (0.48-0.8 us).

RDY (WIRED-AND) AND FLT\_N (WIRED-OR)



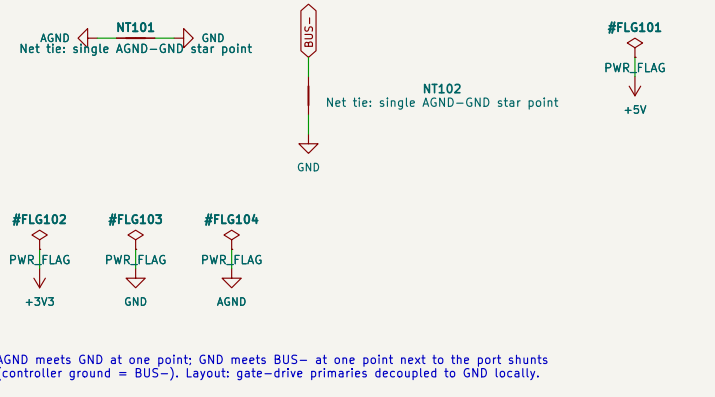
RDY: every NSI6651 RDY (VCC1 and VCC2 UVLO), the +5V and +5V\_GD supervisors and the aux status (live 24 V UV), pulled up HERE to +3V3 (4.99k). FLT\_N: every NSI6651 /FLT + port OC (pull-up on the control board, +5V supervisor TPS3710 107V/20n: RDY low below 4.52-4.63 V (AHCT VCC >= 4.5 V, bias headroom).

PORT STATUS TO PC: HOLD, MOV\_OK, PORT OVER-CURRENT INTO FLT\_N



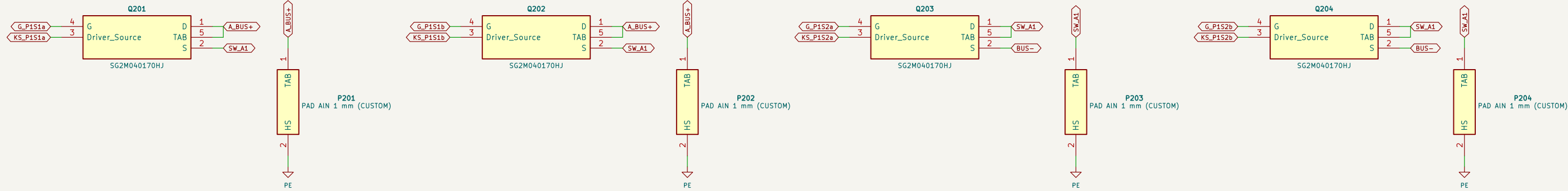
HOLD = A\_HOLD OR B\_HOLD (|I\_port| >= 0.97-1.03 kA, contactor held), MOV\_OK high only while BOTH varistor monitor loops conduct (each reads 'open' below -200 V on its port). Port OC (|I| >= 387-413 A, lean\_shunt) pulls FLT\_N low (open drain): the control board's latch trips without firmware (Rd-fuse gap, port.spec). K\_A / K\_B / K\_PRE pulled low: open = coil off (the hardware interlocks follow on the port sheets).

GROUND STAR POINTS



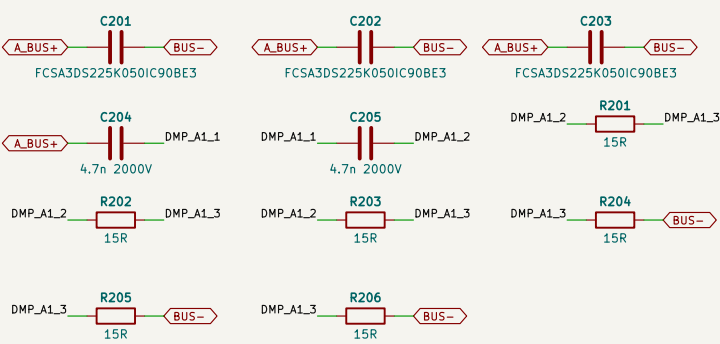
AGND meets GND at one point; GND meets BUS- at one point next to the port shunts (controller ground = BUS-). Layouts gate-drive primaries decoupled to GND locally.

PHASE 1 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



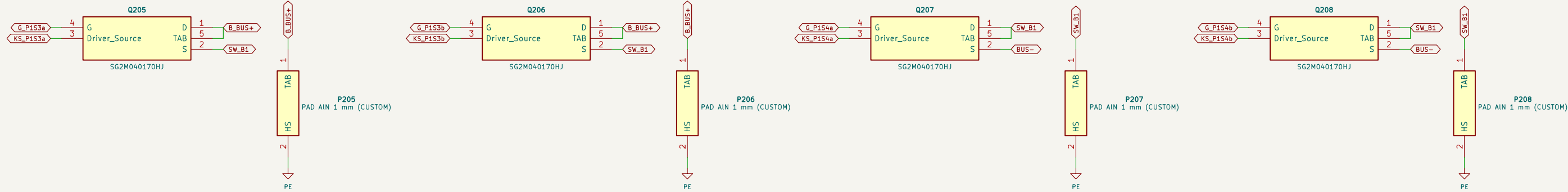
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own  
AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and  
pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 1 LEG A DECOUPLING + RC DAMPER



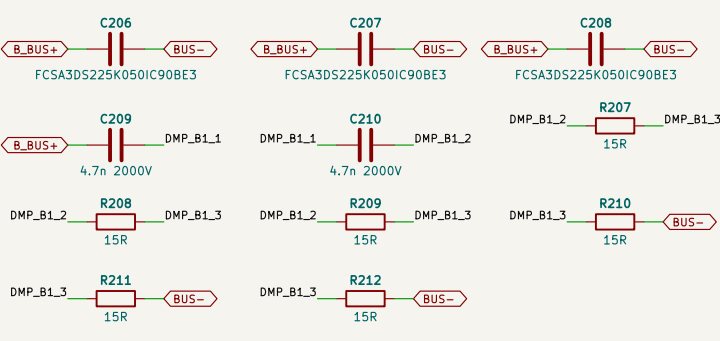
3 x FCSA3D5225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).  
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 1 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



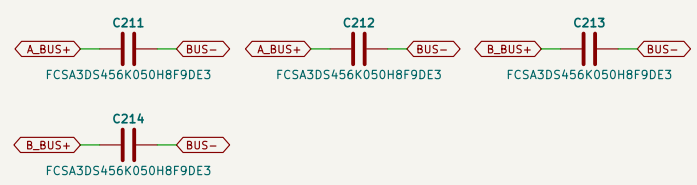
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AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and  
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PHASE 1 LEG B DECOUPLING + RC DAMPER



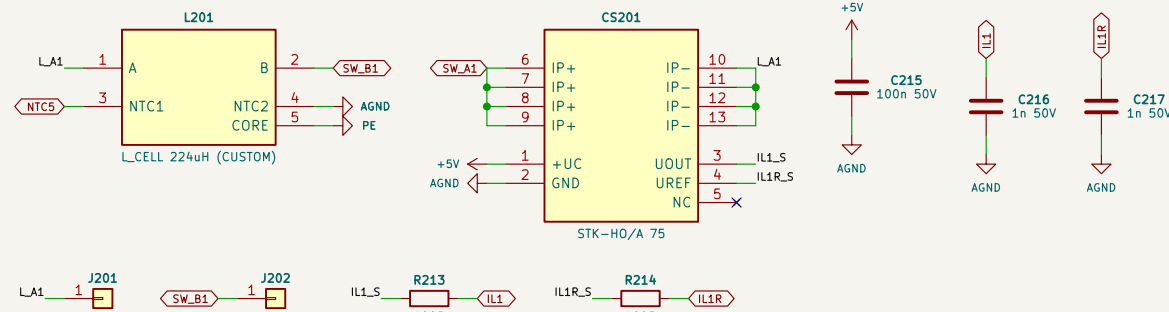
3 x FCSA3D5225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).  
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 1 SHARE OF THE PORT BANKS



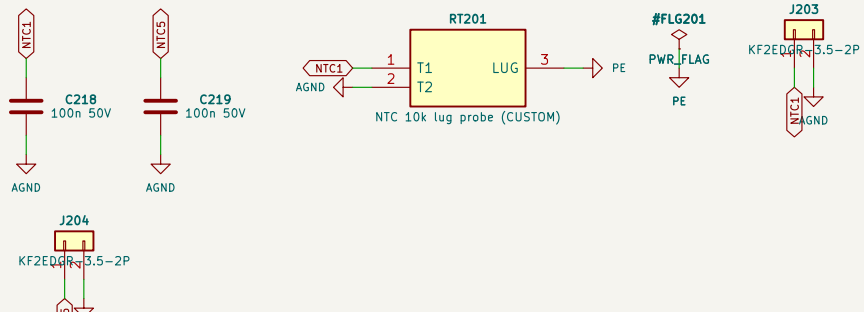
2 x FCSA3D5456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs: the banks of all phases form the  
port A / port B DC links (8 / 8 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 1 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



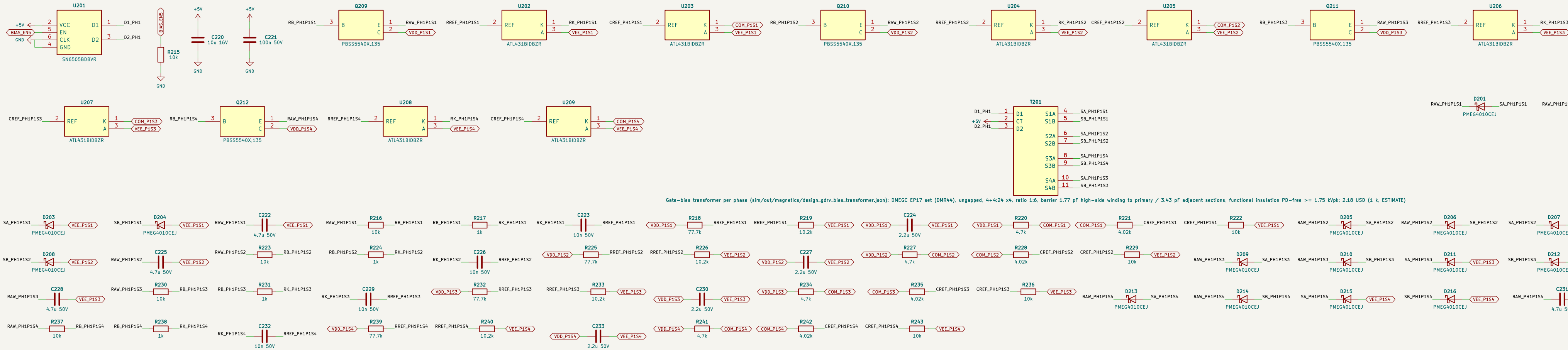
Lead A from SW\_A1 passes the STK-HO/A 75 U-primary (pins 6-9 in, 10-13 out + = SW\_A -> SW\_B) to stud  
LA1: lead B on stud SW\_B1. IL1 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V). 0.5-4.5 V over +/-187.5 A;  
100R +/- 1 mF at the source. IL1R = the sensor's own Uref (pin 4). 100R +/- 1 mF: PV-CTL references its trip  
window to it. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 1 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)

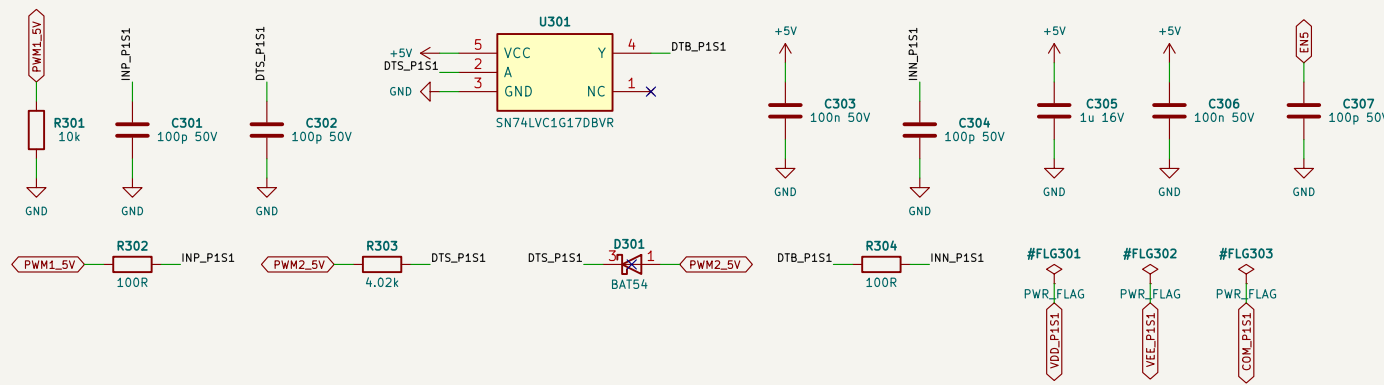


NTC1 = heatsink section of phase 1: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (8-10).  
NTC5 = inductor winding NTC (functional to the winding). 100 mF at the header against switch-node noise.

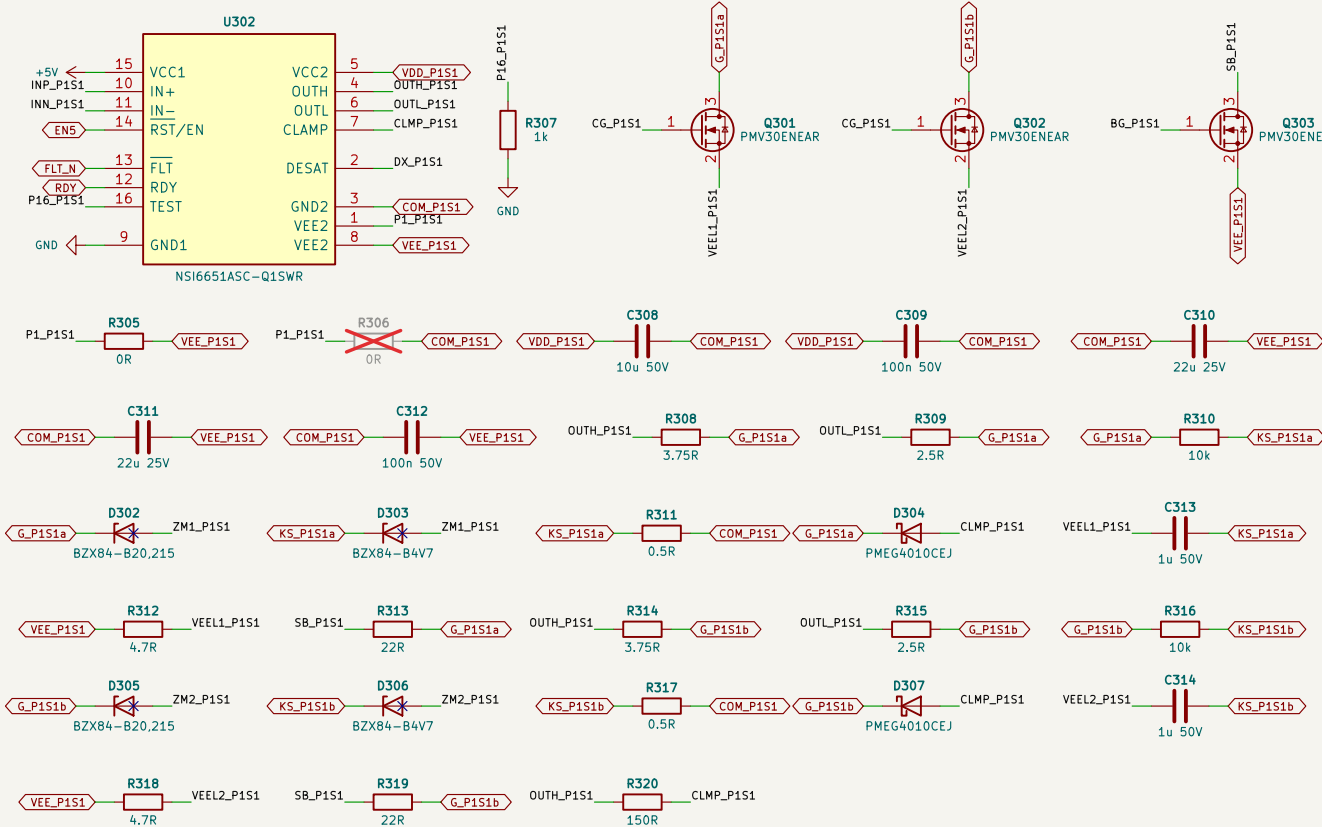
PH1: GATE BIAS (PER PHASE)



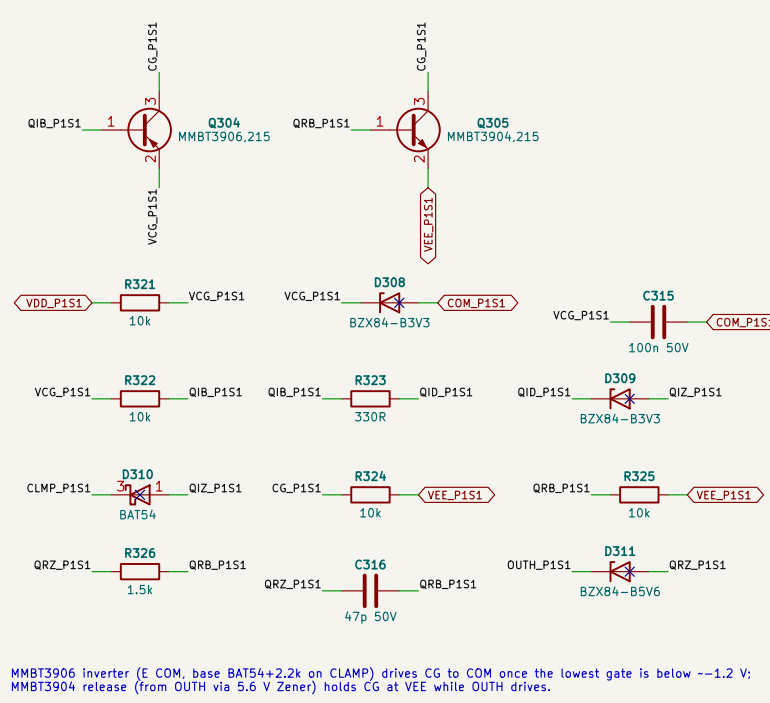
### P1S1: LOGIC SIDE



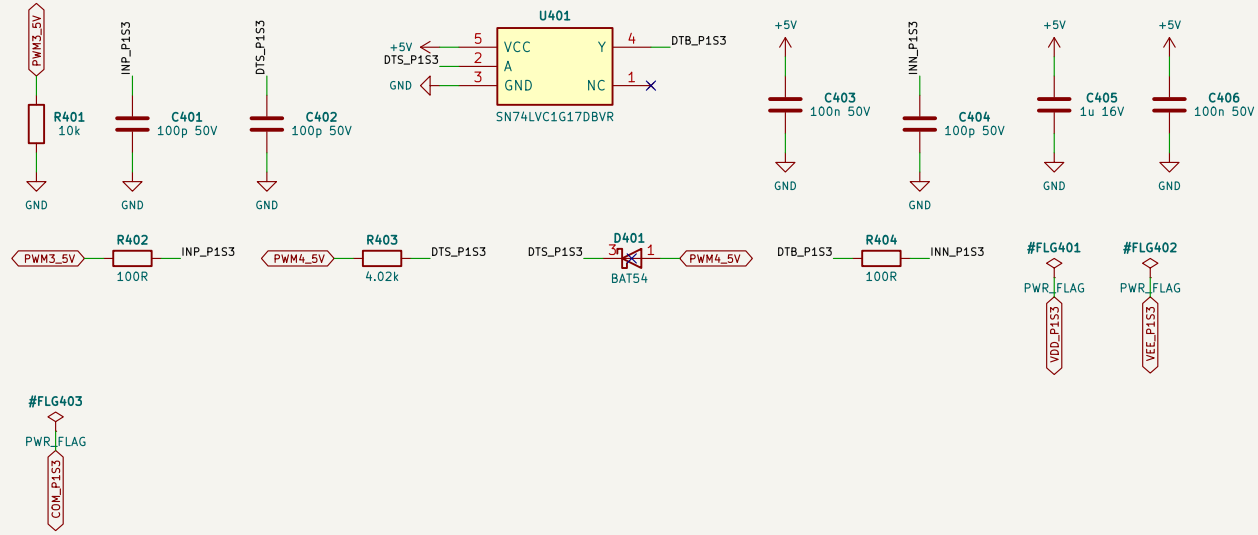
### P1S1: DRIVER AND 2 GATES



### P1S1: MILLER CLAMP CONTROL

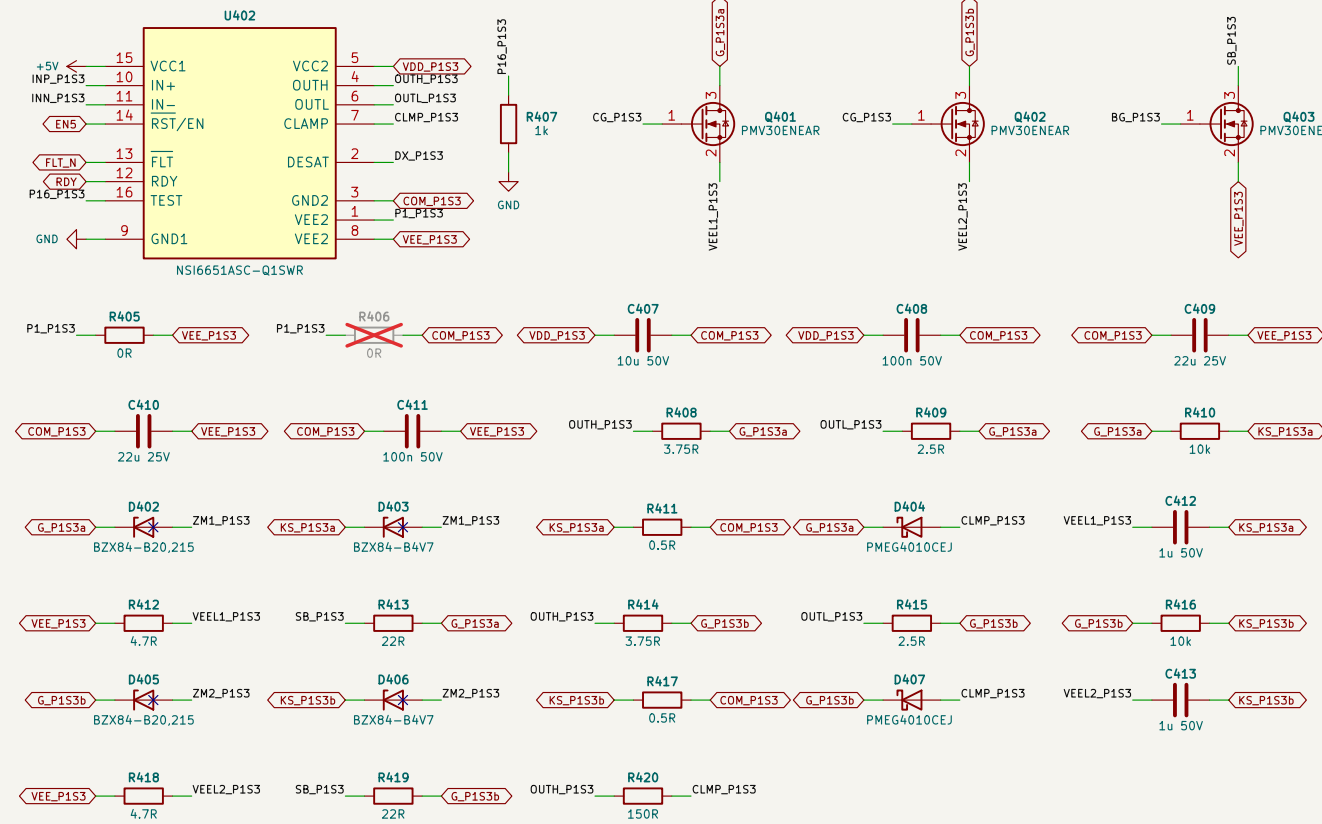


### P1S3: LOGIC SIDE



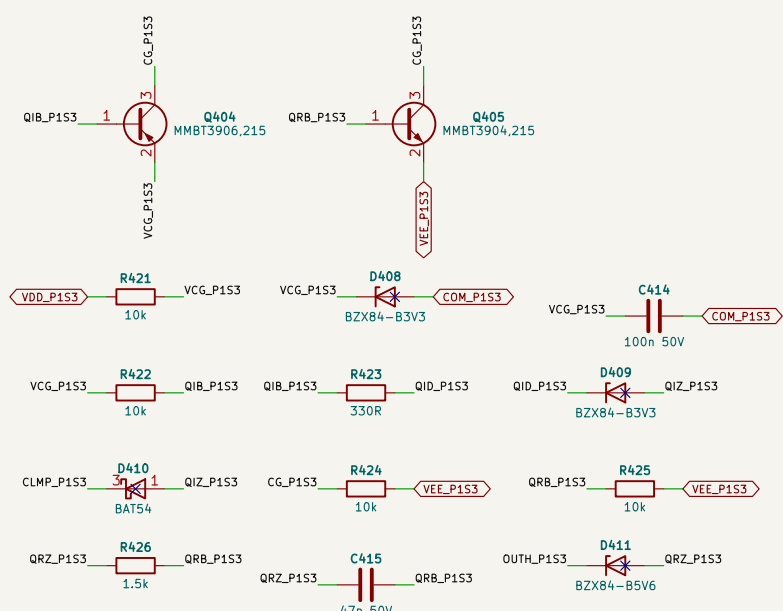
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p  
(Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

### P1S3: DRIVER AND 2 GATES



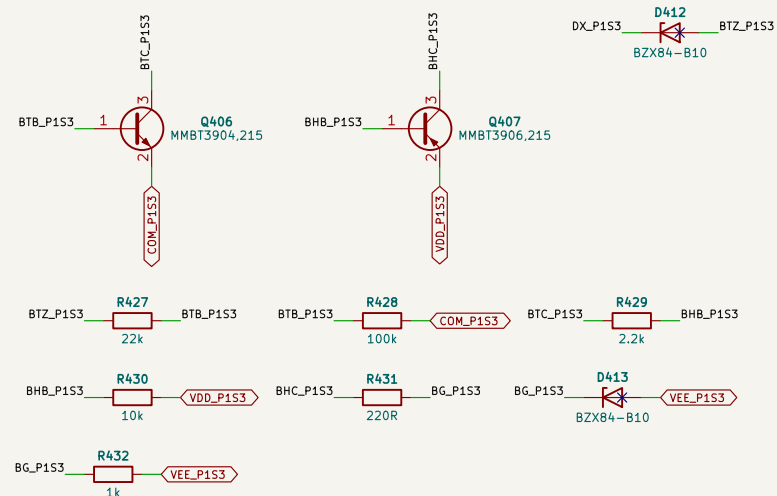
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.  
Miller hold: N-MOSFET clamp (PMV30EENAE) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

### P1S3: MILLER CLAMP CONTROL



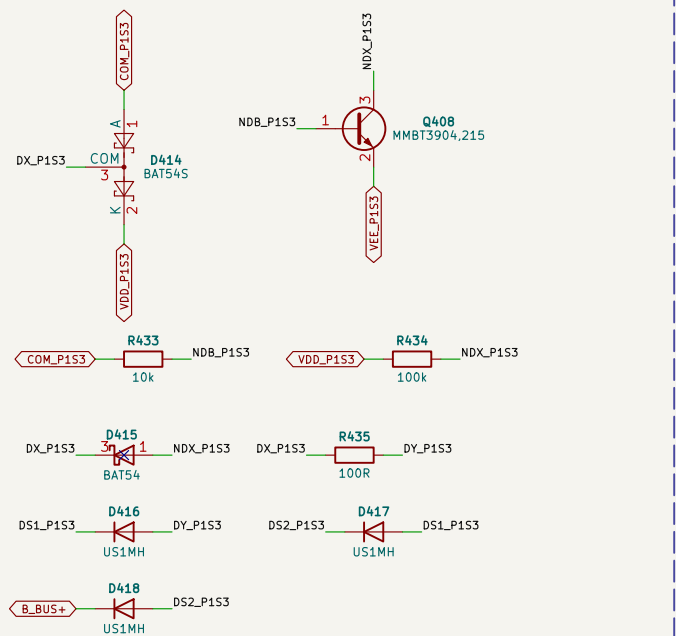
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;  
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

### P1S3: SHORT-CIRCUIT BOOSTER



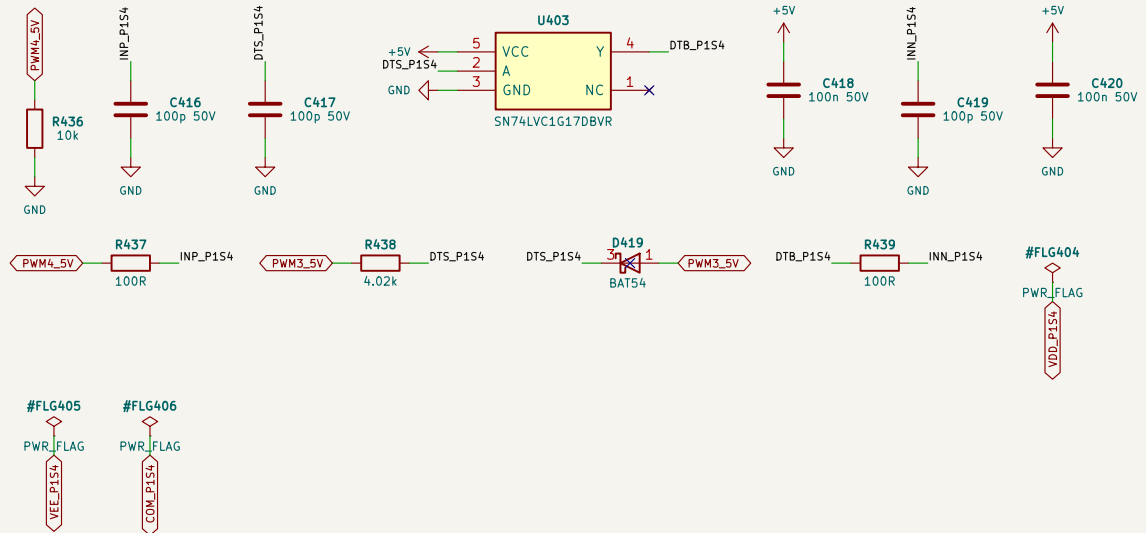
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;  
per device AO3400A + 22R pull the gate to VEE; BG holds -3 us (Lk + FET gates).

### P1S3: DESAT



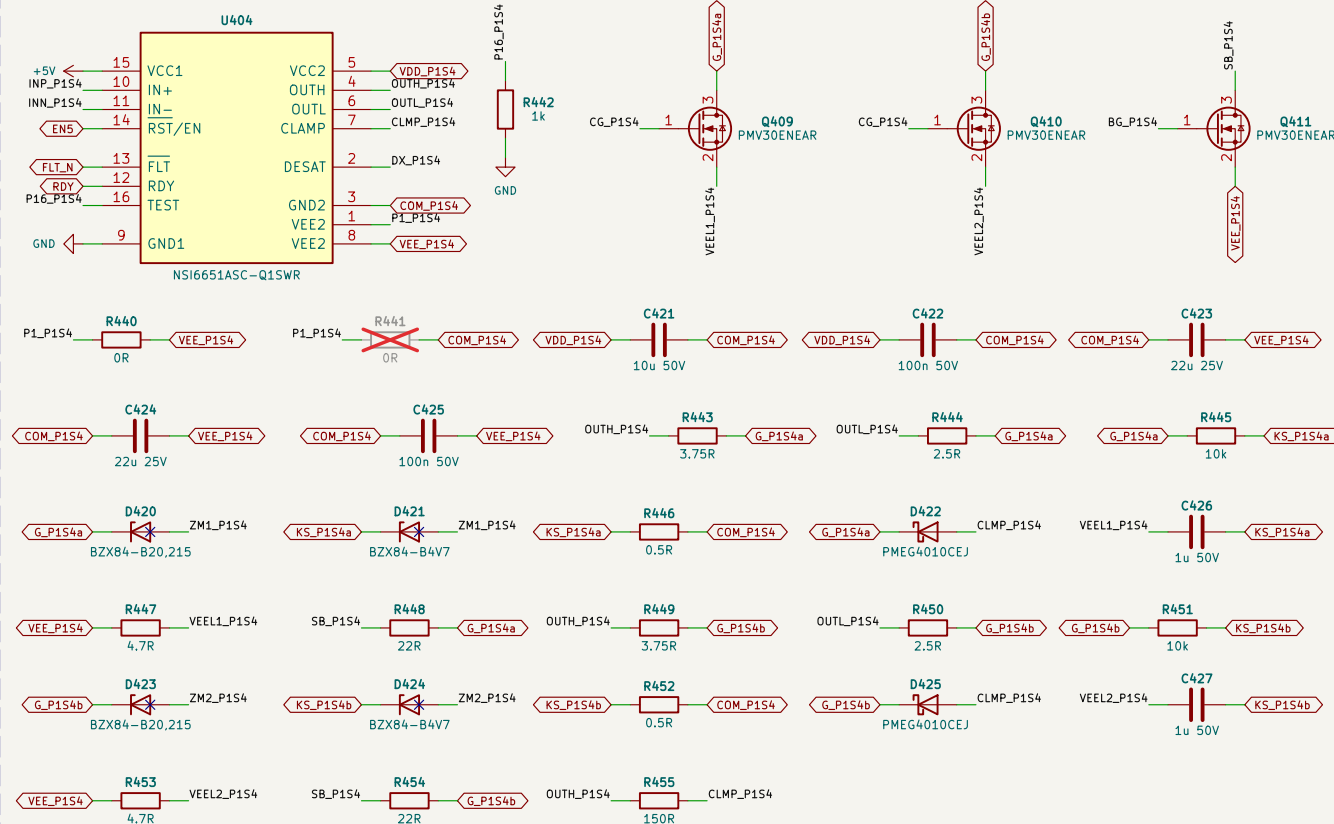
DESAT pin (internal 500 uA blanking source, trip 8.5-10.0 V) - Rk - 3 x US1M - drain.  
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

### P1S4: LOGIC SIDE



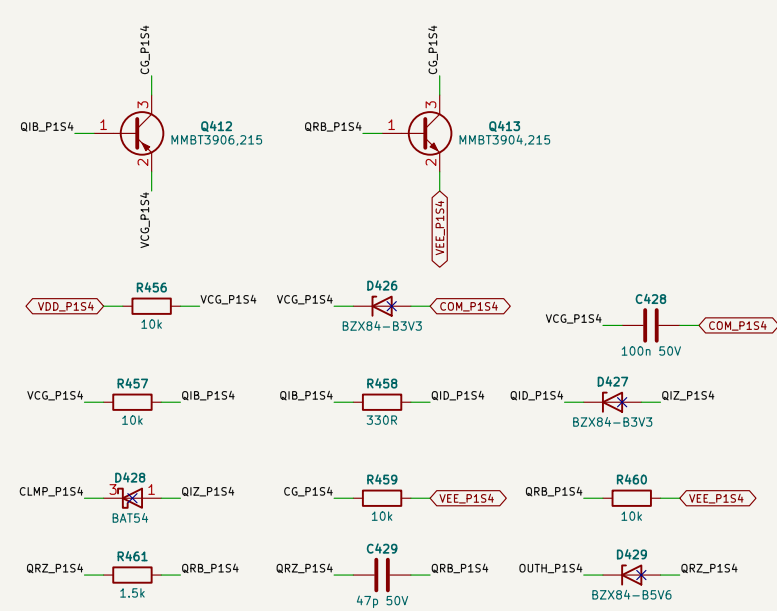
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(Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

### P1S4: DRIVER AND 2 GATES



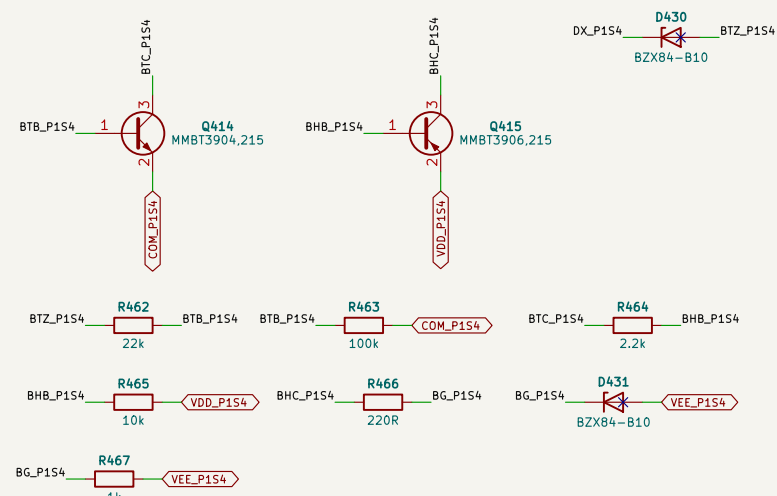
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Miller hold: N-MOSFET clamp (PMV30EENAE) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

### P1S4: MILLER CLAMP CONTROL



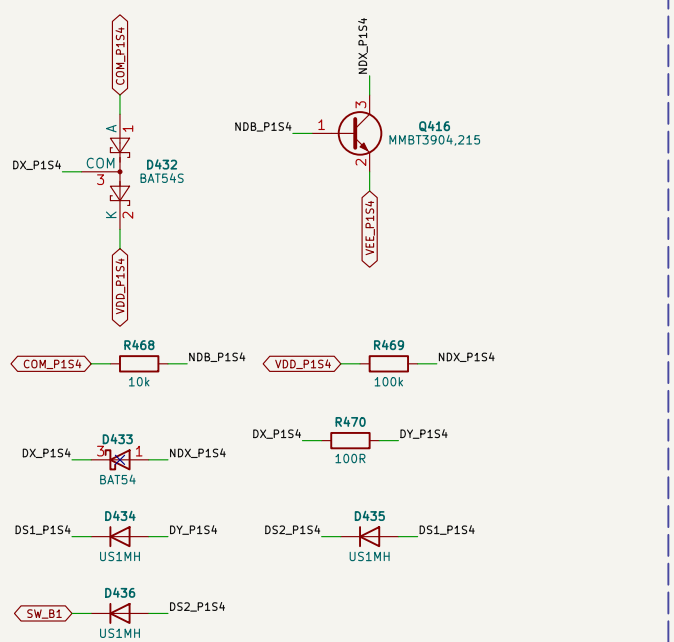
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### P1S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;  
per device AO3400A + 22R pull the gate to VEE; BG holds -3 us (Lk + FET gates).

### P1S4: DESAT



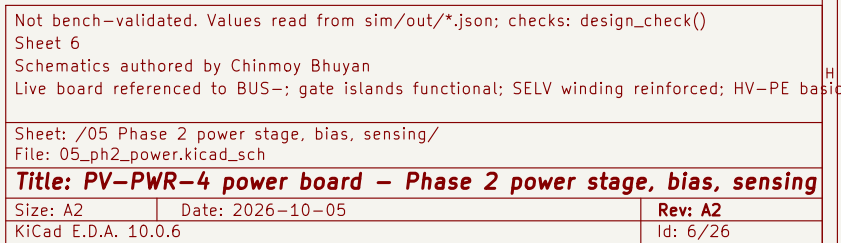
DESAT pin (internal 500 uA blanking source, trip 8.5-10.0 V) - Rk - 3 x US1M - drain.  
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

### LEG B: LAYOUT RULES

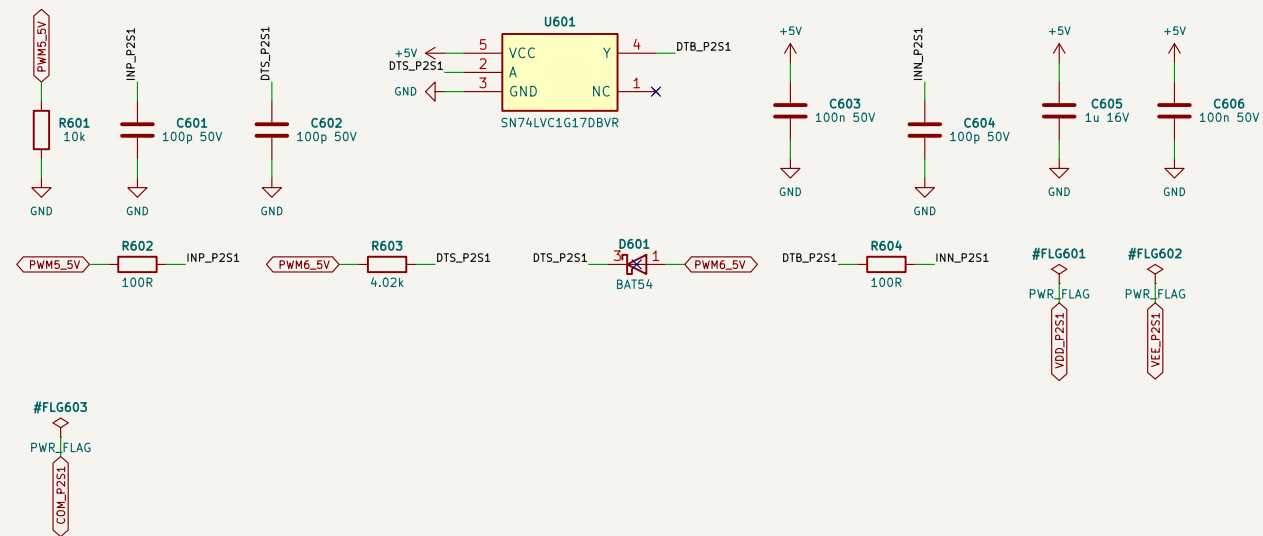
Gate and Kelvin of each device as coupled pairs of equal length (+/- 10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C\_VL at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string l = 2.5 mm pad gap per diode; board coated to PDI (IC-22). Miller hold (nspice, gdrv); die +1.41 V vs VGS(on) min 1.94 V at 1/5 - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

PHASE 2 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH

The schematic diagram illustrates the internal components and connections of the L501 and C5501 modules. The L501 module (left) features a NTCT1 core with pins A, B, C, D, E, and F. It is connected to LA2 (pin 1), SW\_B2 (pin 2), AGND (pin 4), and PE (pin 5). The C5501 module (right) features a STK-HQ/A 75 core with pins 1 through 13. It is connected to LA2 (pin 10), SW\_A2 (pin 6), SW\_B2 (pin 7), AGND (pin 8), +5V (pin 9), +5V (pin 1), UIC (pin 2), GND (pin 3), UOUT (pin 4), UREF (pin 5), IL2\_S (pin 11), IL2\_S (pin 12), and IL2\_S (pin 13). The diagram also shows the internal components of the modules, including J501, J502, R513, R514, C515, C516, and C517.

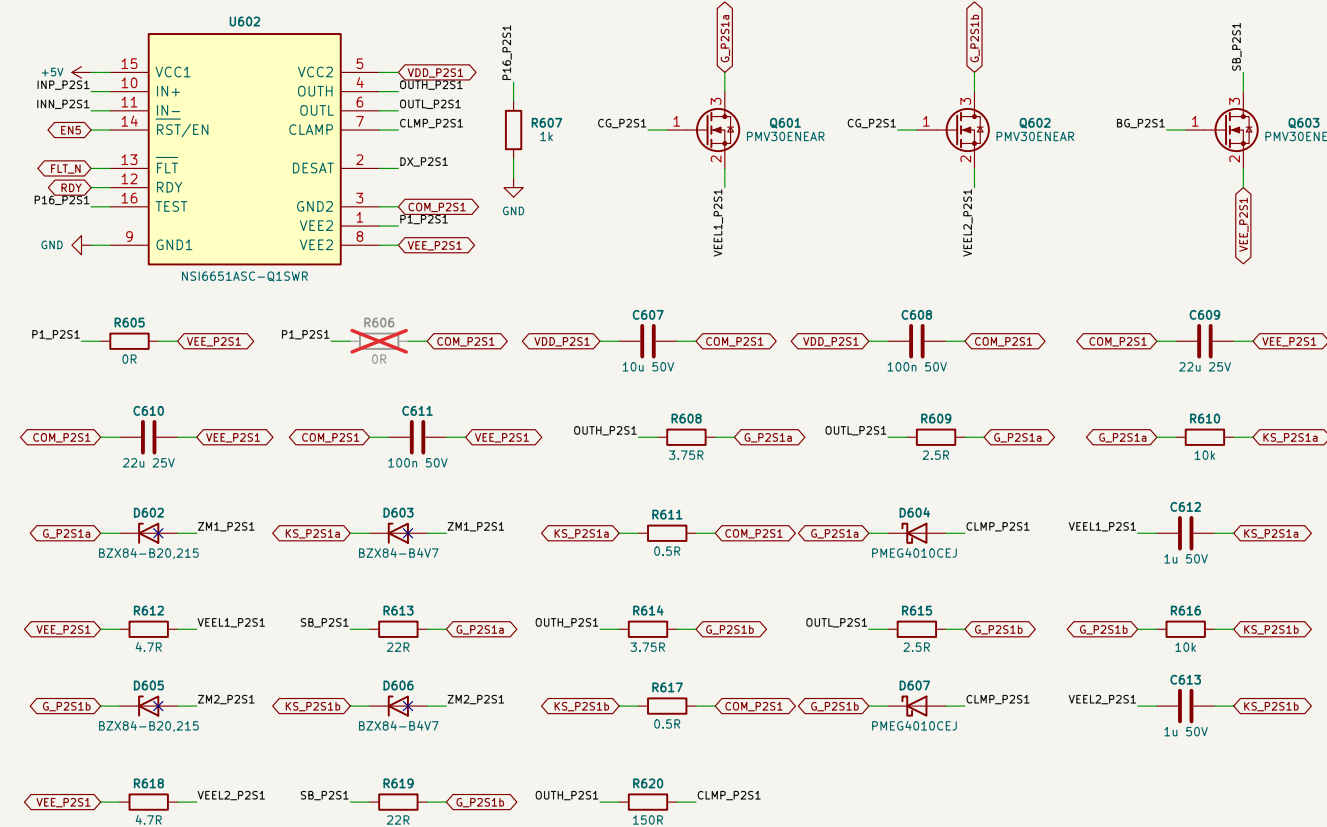
[illegible]

## P2S1: LOGIC SIDE



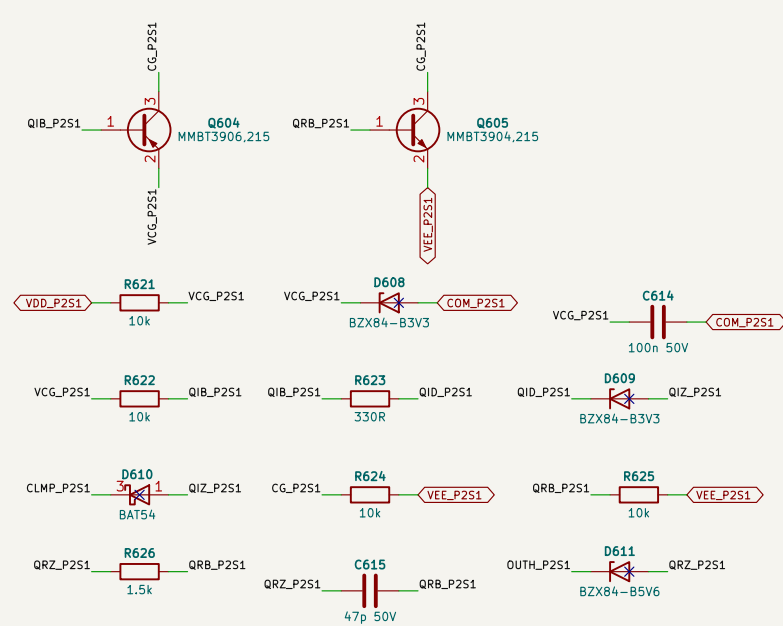
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## P2S1: DRIVER AND 2 GATES



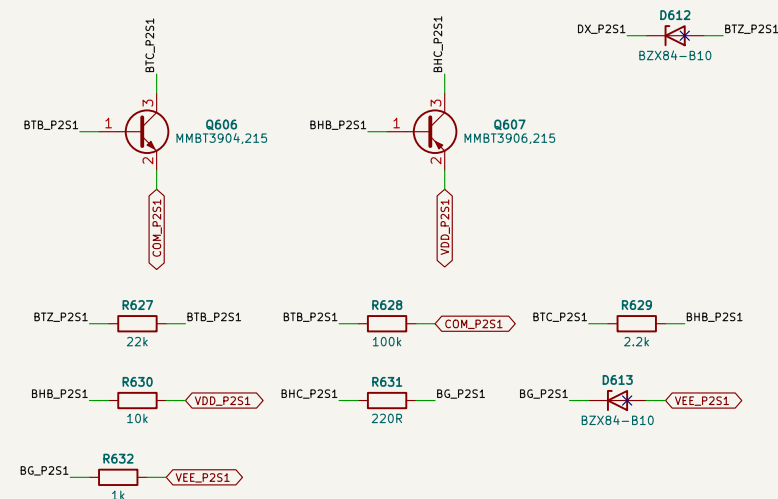
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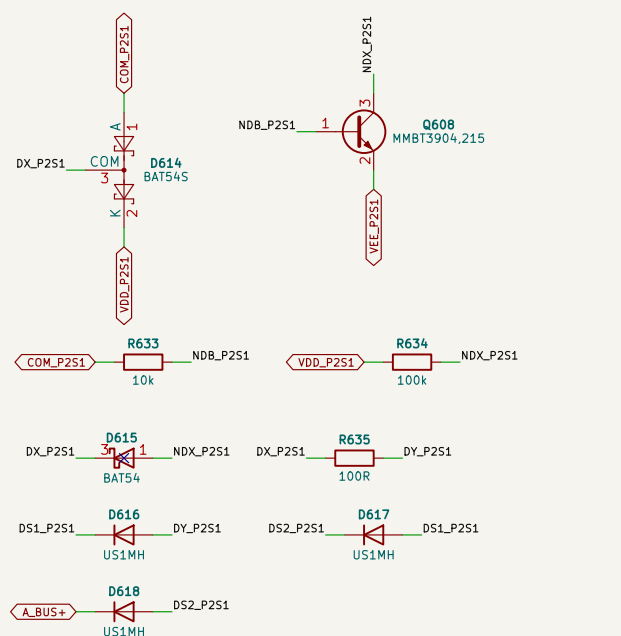
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## P2S1: SHORT-CIRCUIT BOOSTER



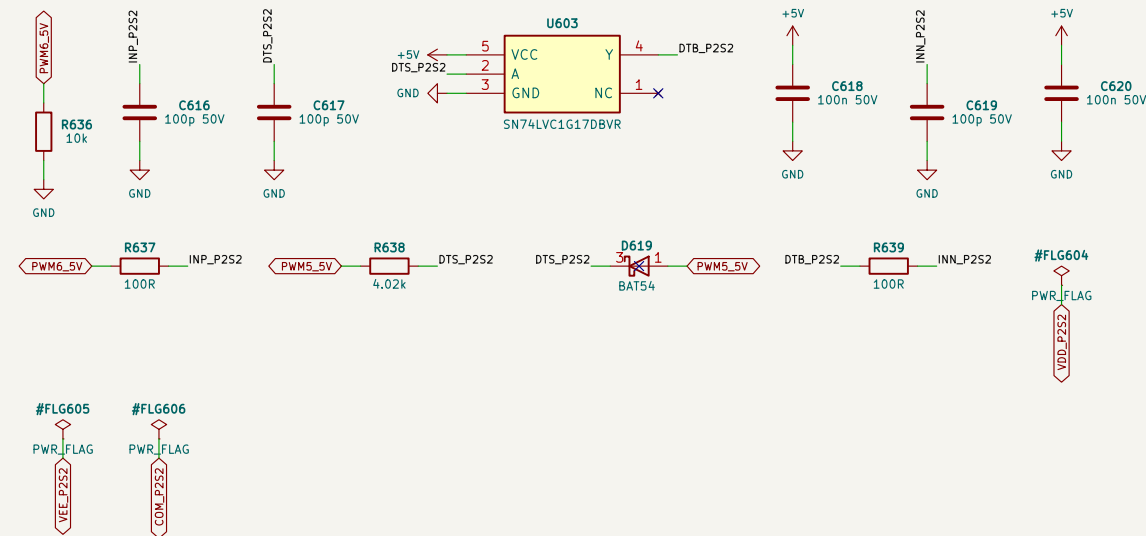
DESAT > 10.3–10.9 V (above the 10.0 V maximum trip of the –Q1 grade) → MMBT3904 → MMBT3906 charges BG to VEE + 10 V; per device AQ3400A + 22R pull the gate to VEE; BG holds ~3  $\mu$ s (1k + FET gates).

P2S1: DESAT



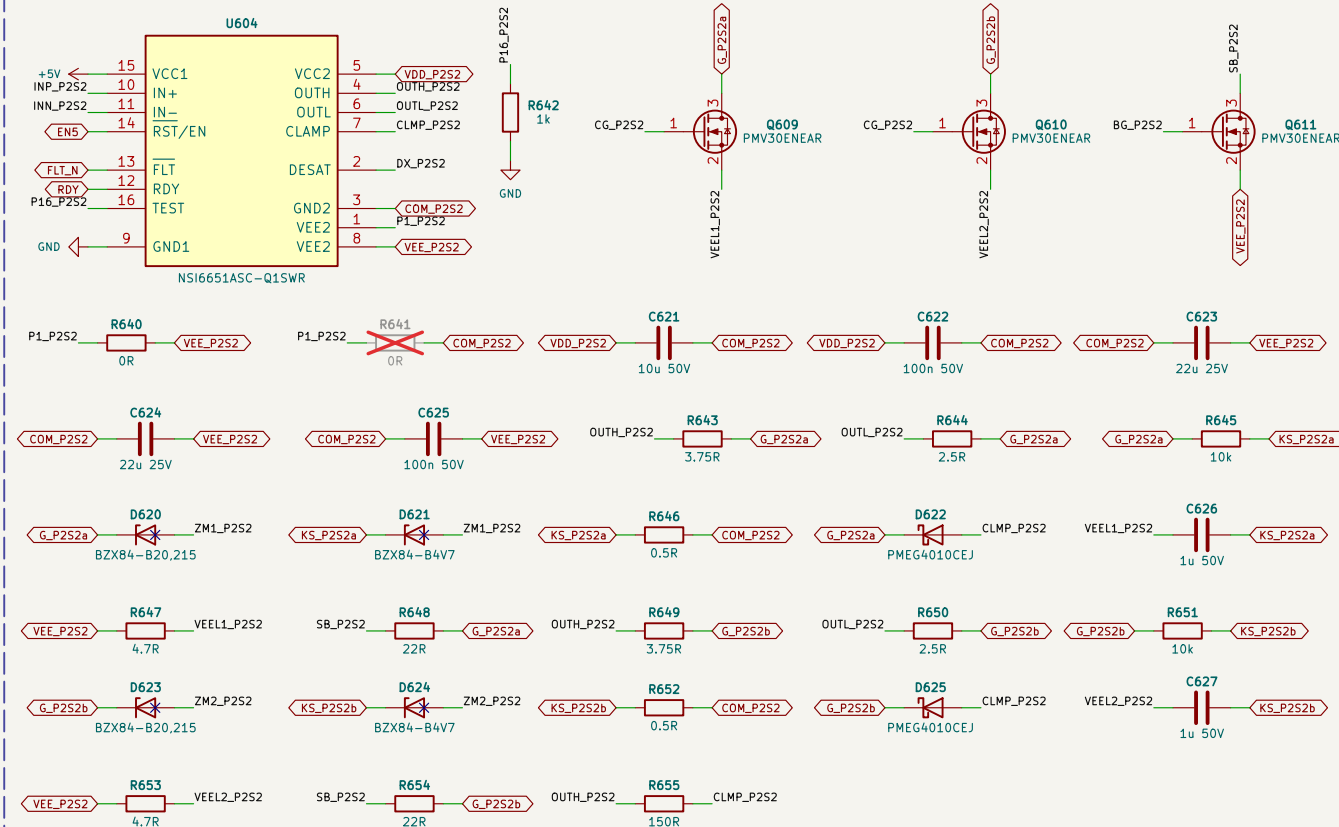
DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 3 x 10M – drain.  
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

## P2S2: LOGIC SIDE



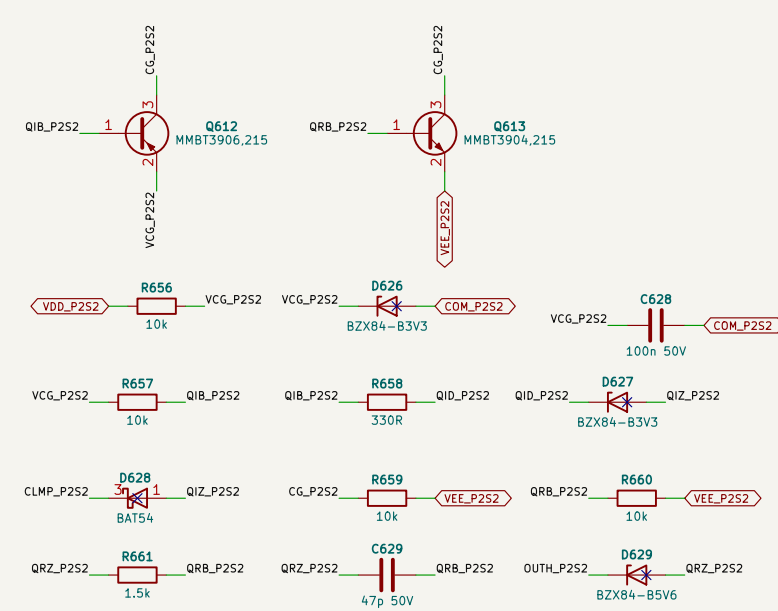
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P2S2: DRIVER AND 2 GATES



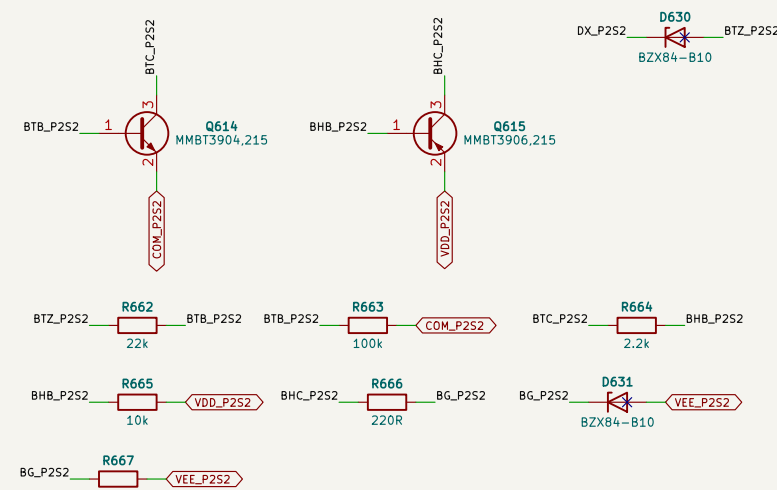
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.  
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

## P2S2: MILLER CLAMP CONTROL



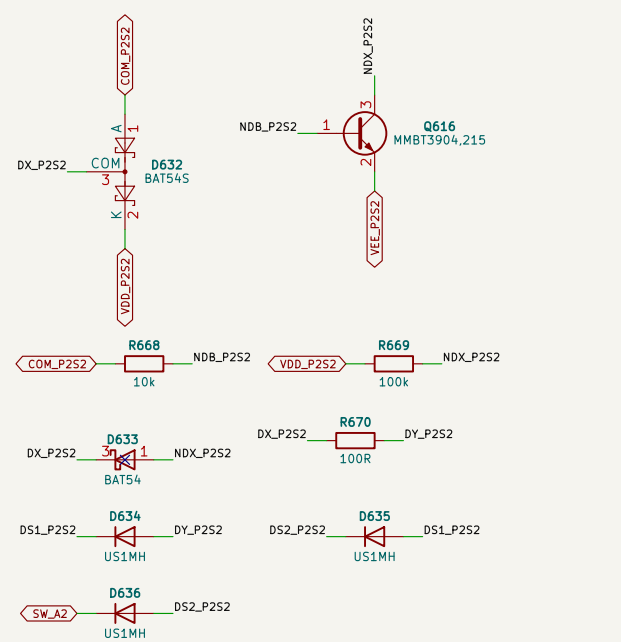
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below  $-1.2$  V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

## P2S2: SHORT-CIRCUIT BOOSTER



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P2S2: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 3 x US1M – drain.  
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

## LEG A: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star: Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C<sub>VL</sub> at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (ngspice, gdrv): die +1.41 V vs VGS(th) min 1.94 V at 175 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

Not bench-validated. Values read from sim/out/\*.json; checks: design\_check()  
Sheet 7  
Schematics authored by Chinmoy Bhuyan  
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /06 Phase 2 leg A gate drive (S1, S2)/  
File: 06\_ph2\_gda.kicad\_sch

Title: PV-PWR-4 power board - Phase 2 leg A gate drive (S1, S2)

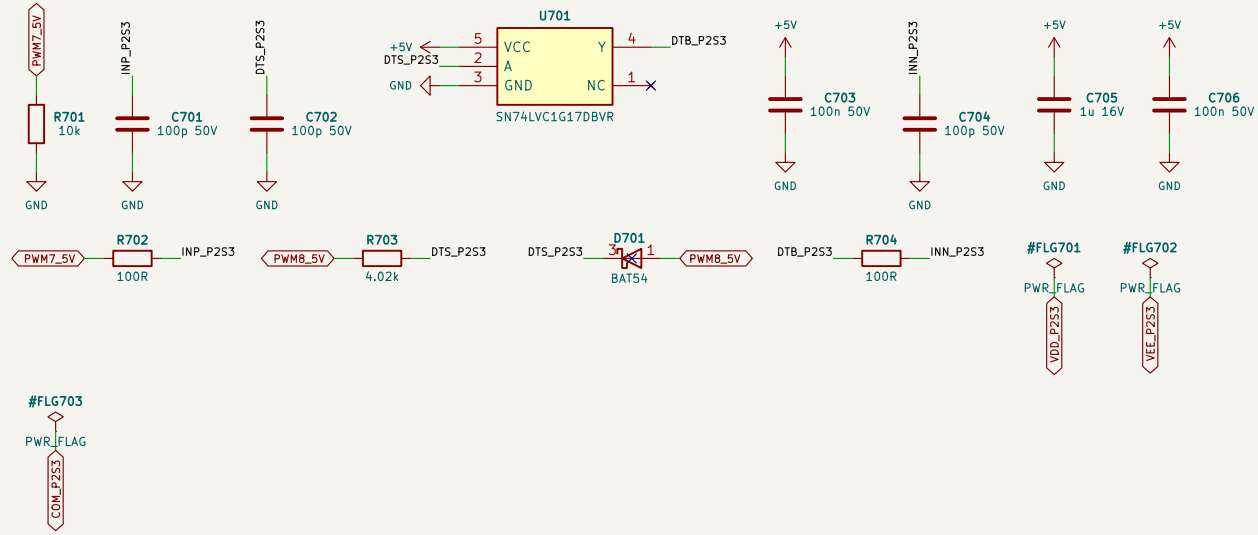
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| Size: A2 | Date: 2026-10-05 |
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Rev: A2

KiCad E.D.A. 10.0.6

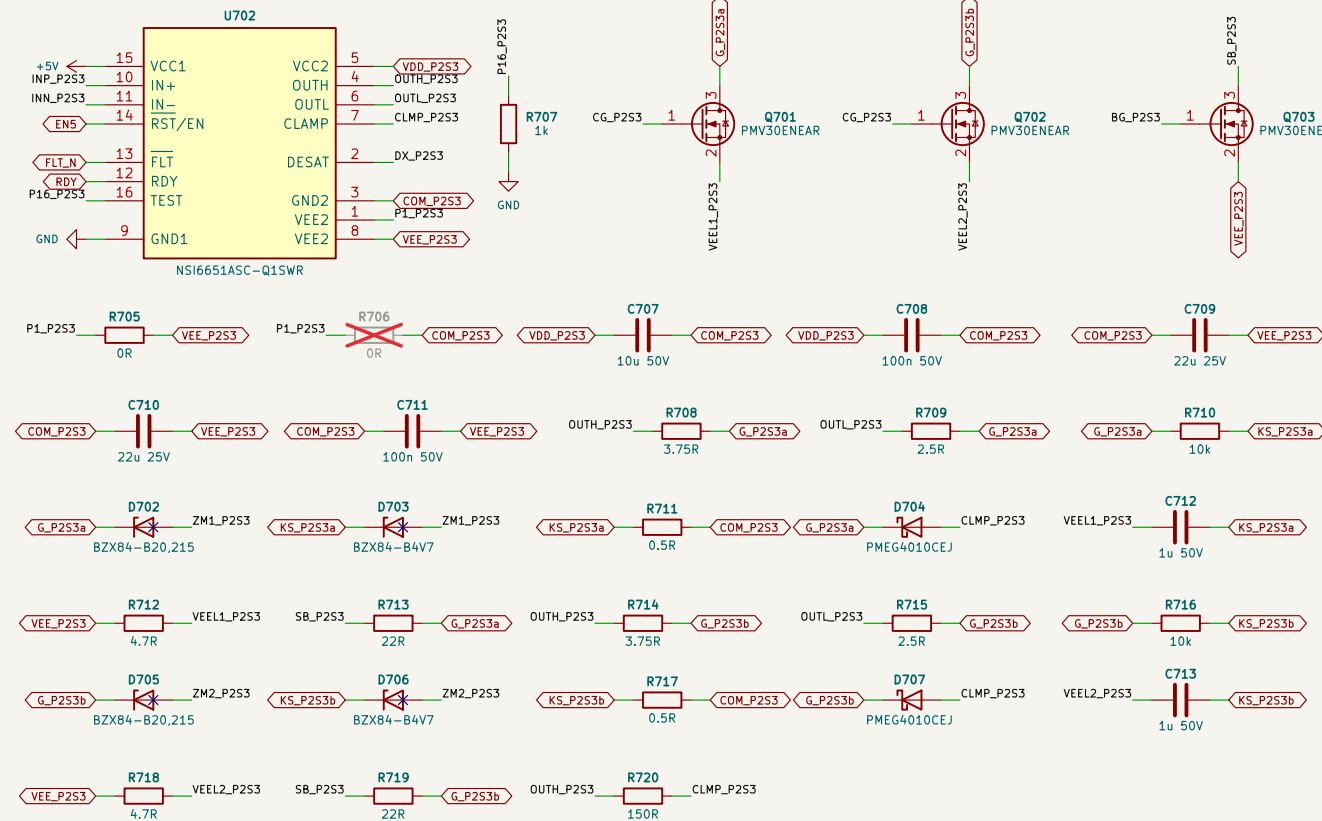
Id: 7/26

### P2S3: LOGIC SIDE



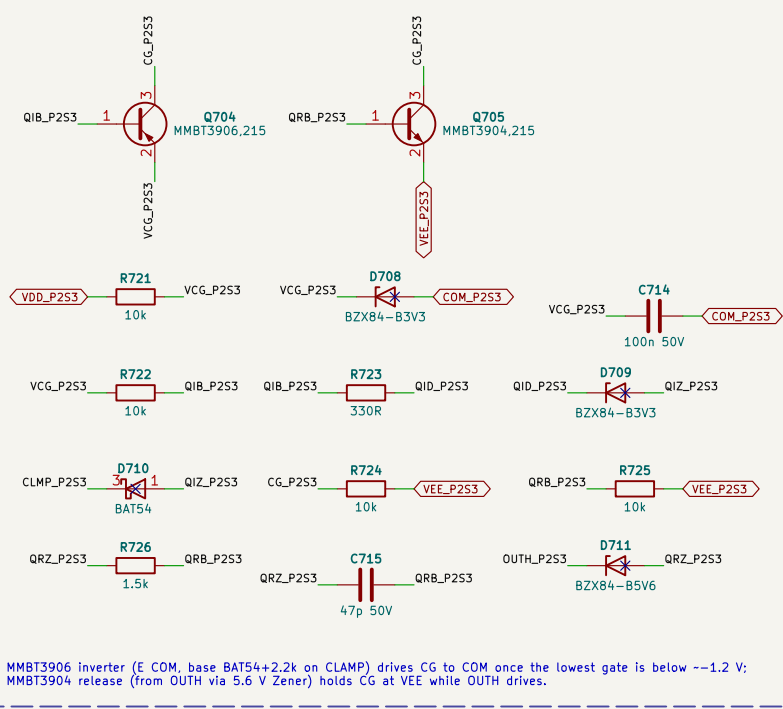
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p  
(Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

### P2S3: DRIVER AND 2 GATES



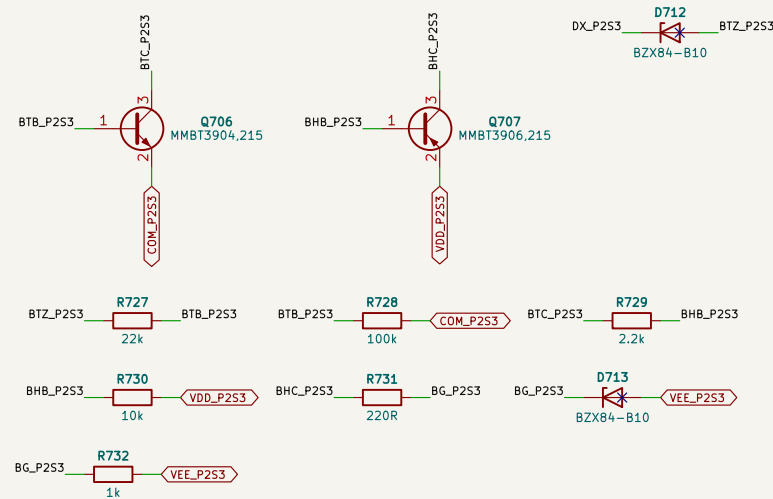
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.  
Miller hold: N-MOSFET clamp (PMV30EEN) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

### P2S3: MILLER CLAMP CONTROL



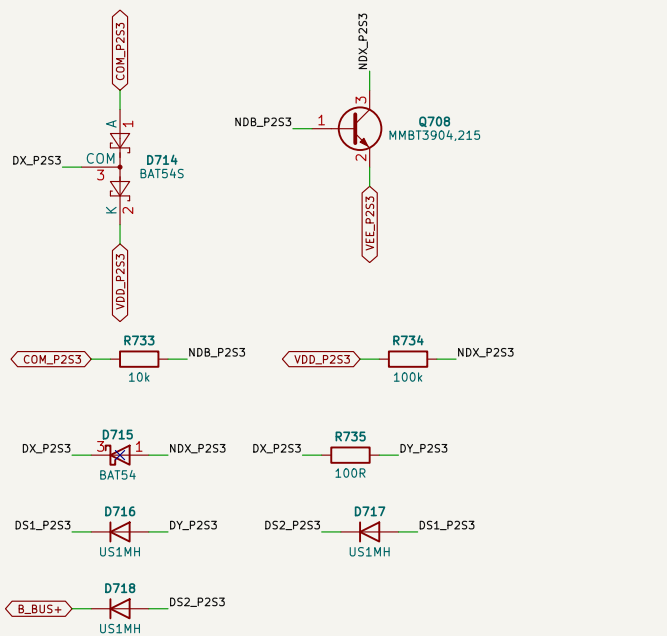
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### P2S3: SHORT-CIRCUIT BOOSTER



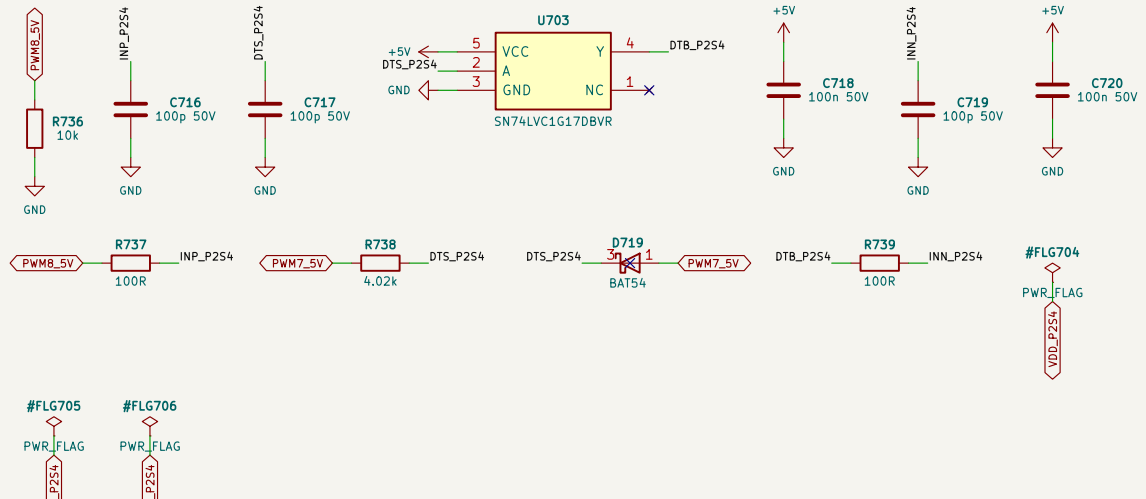
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per device AQ3400A + 22R pull the gate to VEE; BG holds ~3 us (Lk + FET gates).

### P2S3: DESAT



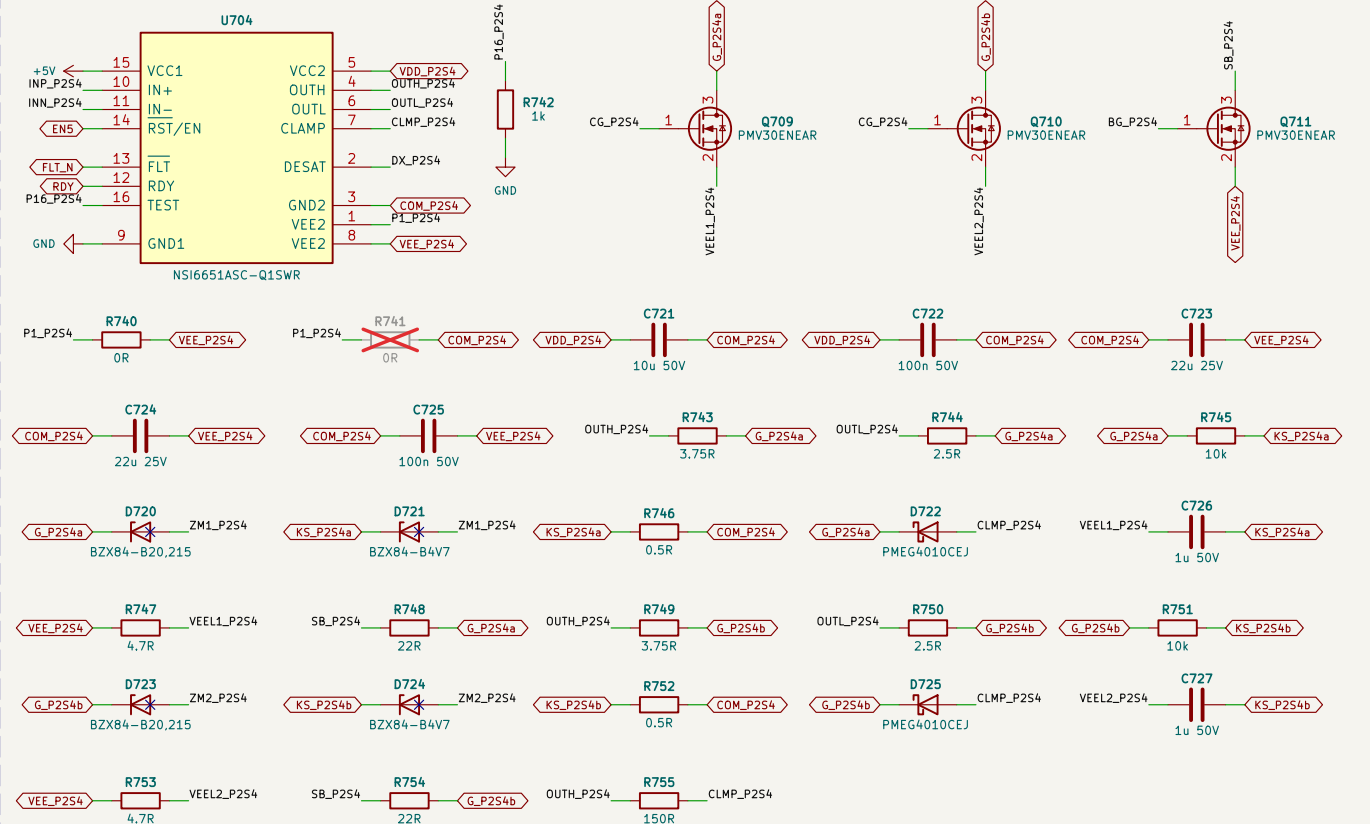
DESAT pin (Internal 500 uA blanking source, trip 8.5-10.0 V) -> Rk - 3 x US1M - drain.  
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

### P2S4: LOGIC SIDE



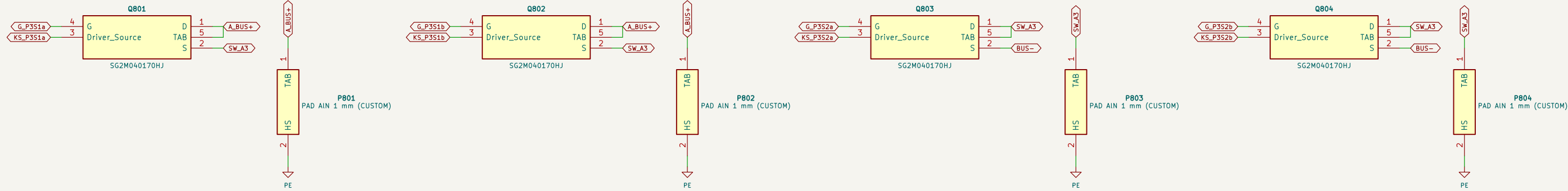
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p  
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### P2S4: DRIVER AND 2 GATES



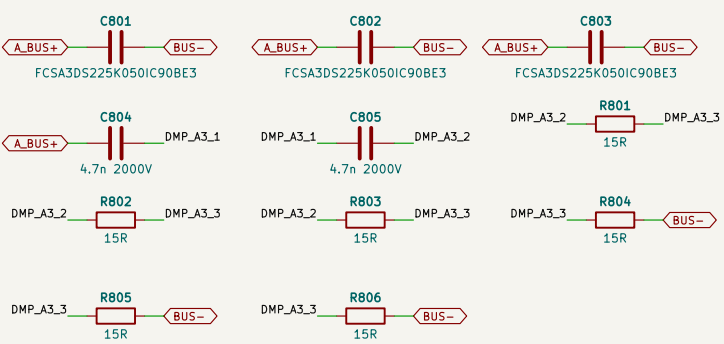
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.  
Miller hold: N-MOSFET clamp (PMV30EEN) gate-pin to a

PHASE 3 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



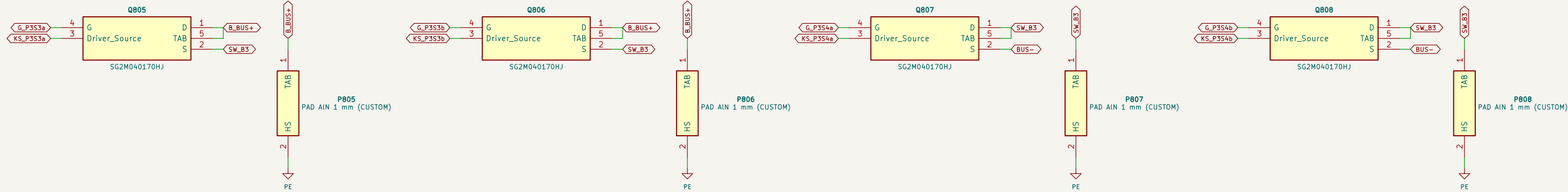
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own  
AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and  
pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 3 LEG A DECOUPLING + RC DAMPER



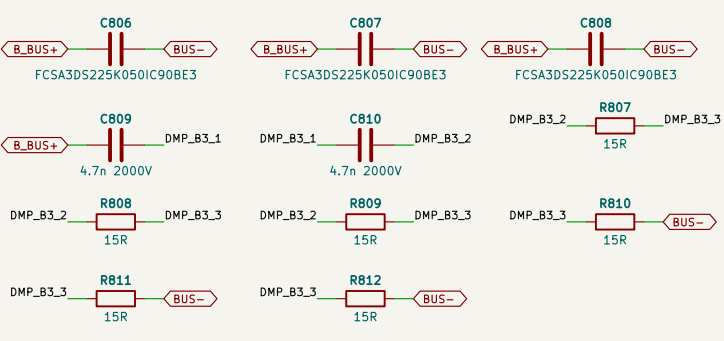
3 x FCSA3D5225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).  
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 3 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



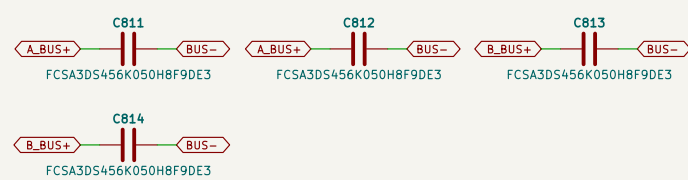
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PHASE 3 LEG B DECOUPLING + RC DAMPER



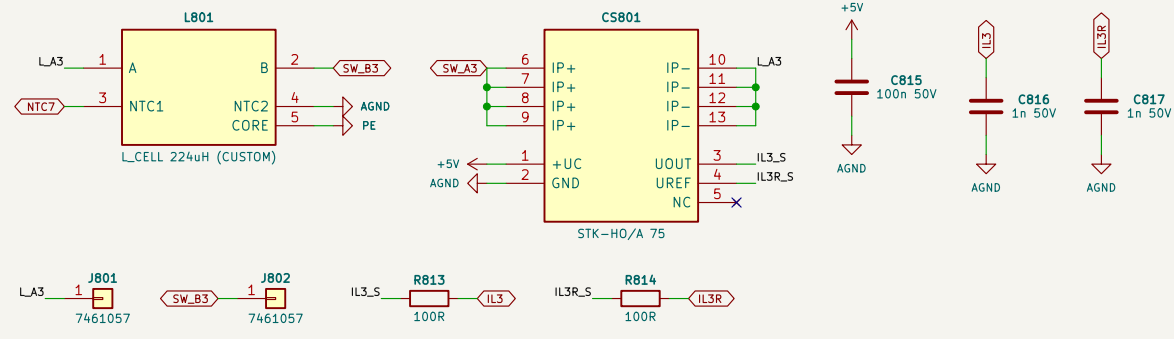
3 x FCSA3D5225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).  
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 3 SHARE OF THE PORT BANKS



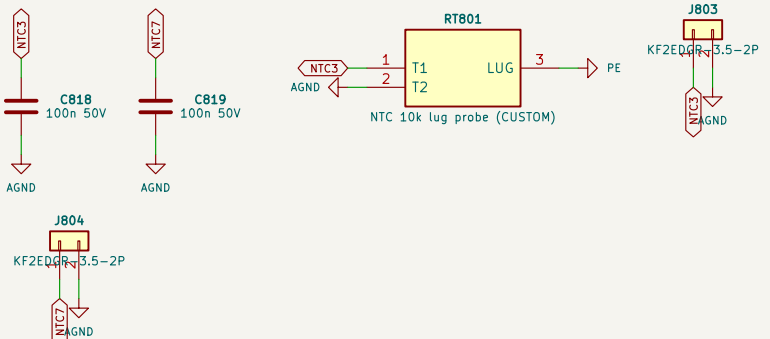
2 x FCSA3D5456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs; the banks of all phases form the  
port A / port B DC links (8 / 8 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 3 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



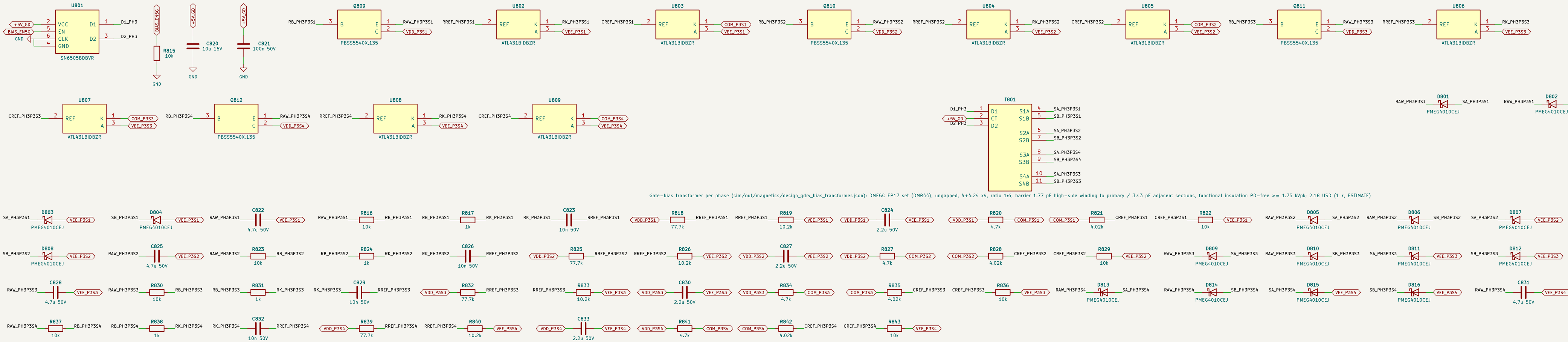
Lead A from SW\_A3 passes the STK-HQ/A 75 U-primary (pins 6-9 in, 10-13 out) + = SW\_A -> SW\_B to stud  
LA3: lead B on stud SW\_B3. IL3 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V). 0.5-4.5 V over +/-187.5 A;  
100R +/- 1 mV at the source. IL3R = the sensor's own Uref (pin 4). 100R +/- 1 mV: PV-CTL references its trip  
window to it. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 3 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



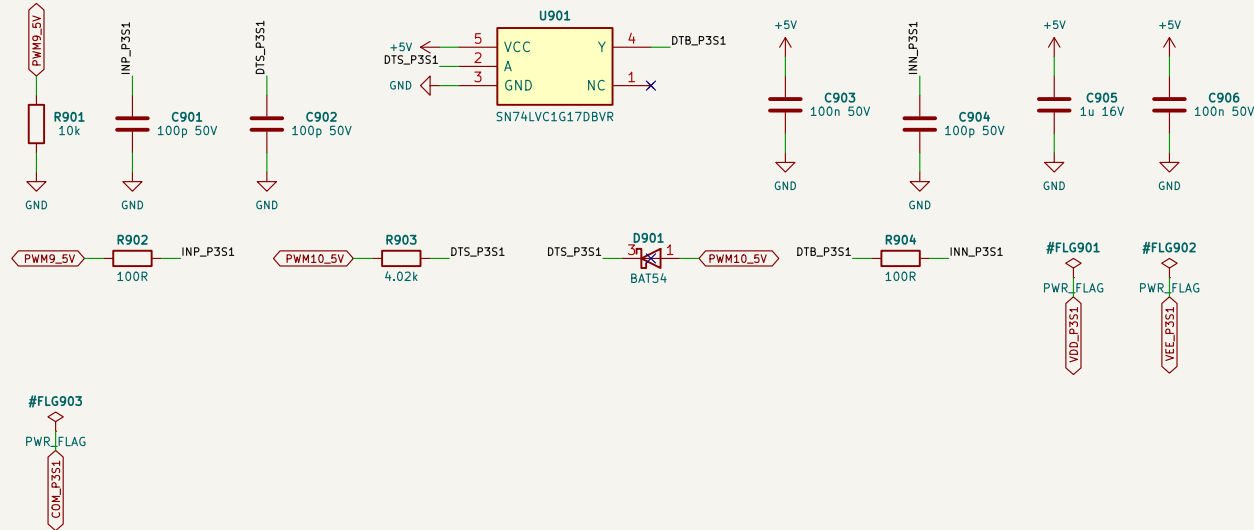
NTC3 = heatsink section of phase 3: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (8-10).  
NTC7 = inductor winding NTC (functional to the winding). 100 mV at the header against switch-node noise.

PH3: GATE BIAS (PER PHASE)



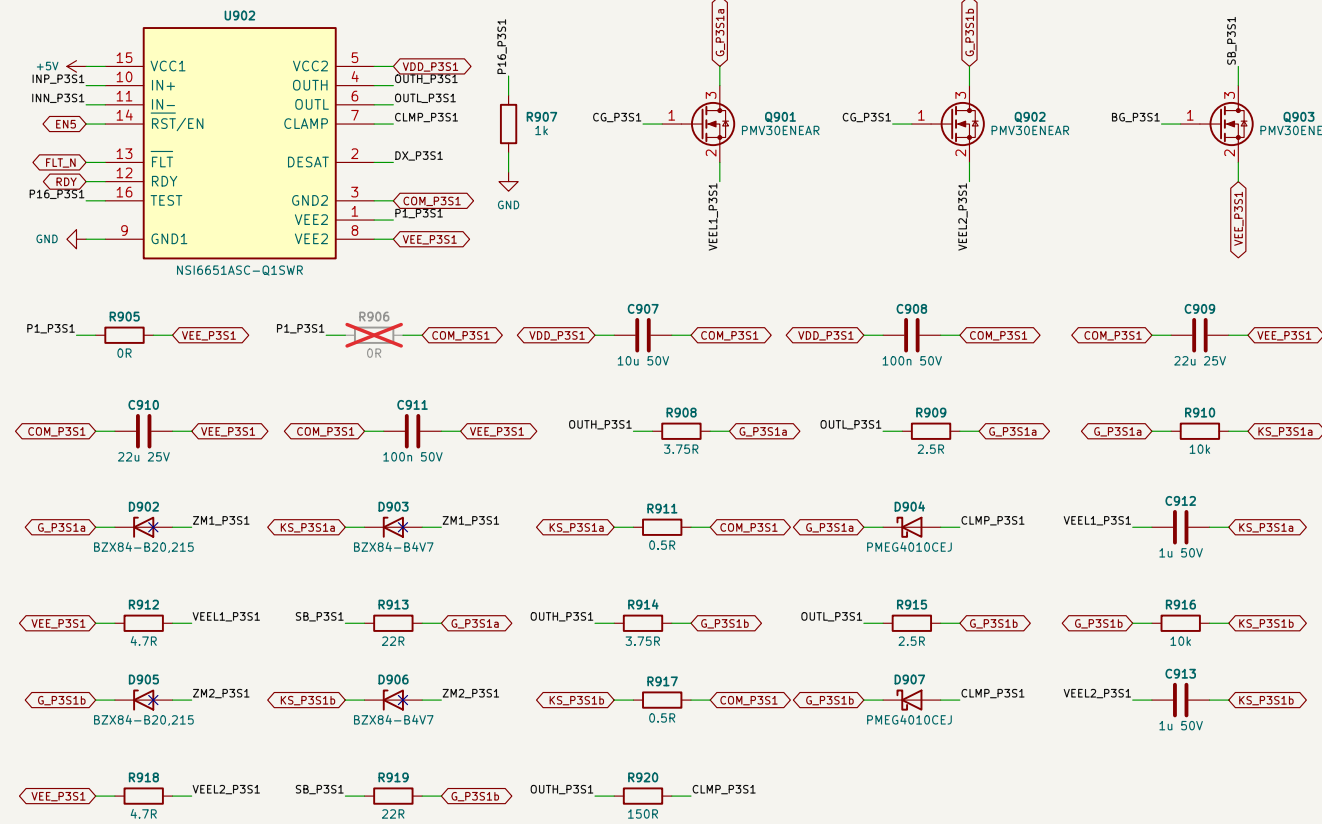
SN6505B push-pull 420 kHz from +5V\_GD, BIAS\_EN high = on; T\_BIAS4 custom, functional insulation  
(PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

### P3S1: LOGIC SIDE



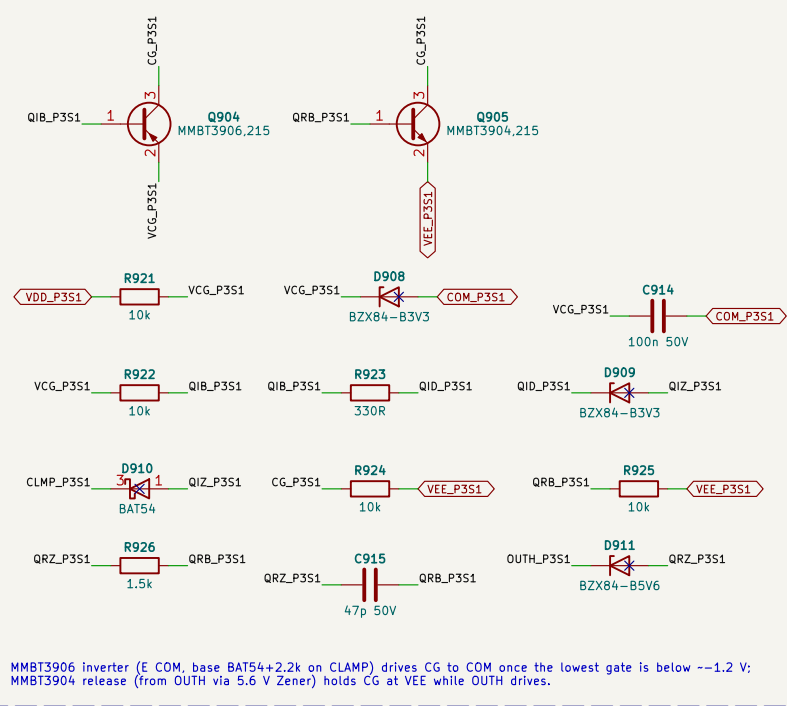
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(Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

### P3S1: DRIVER AND 2 GATES



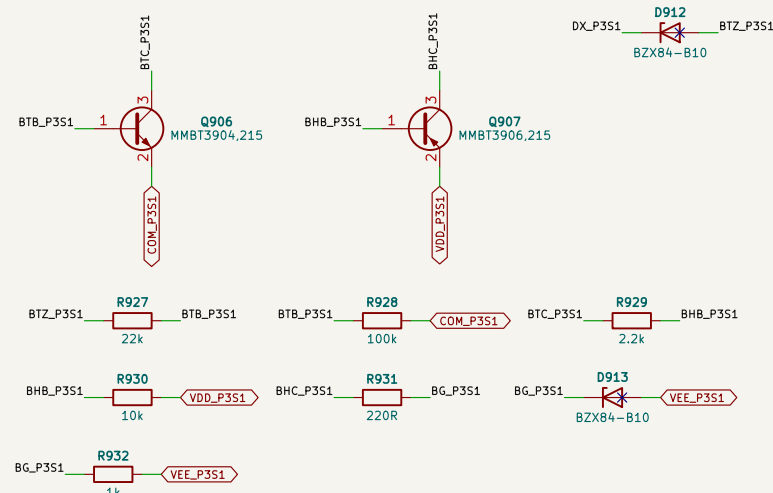
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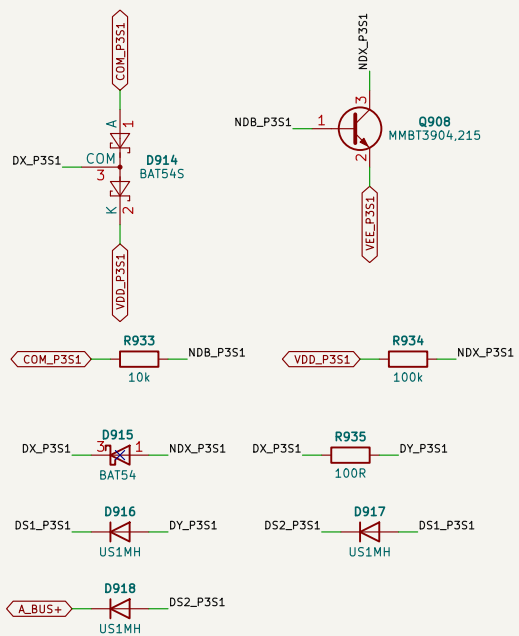
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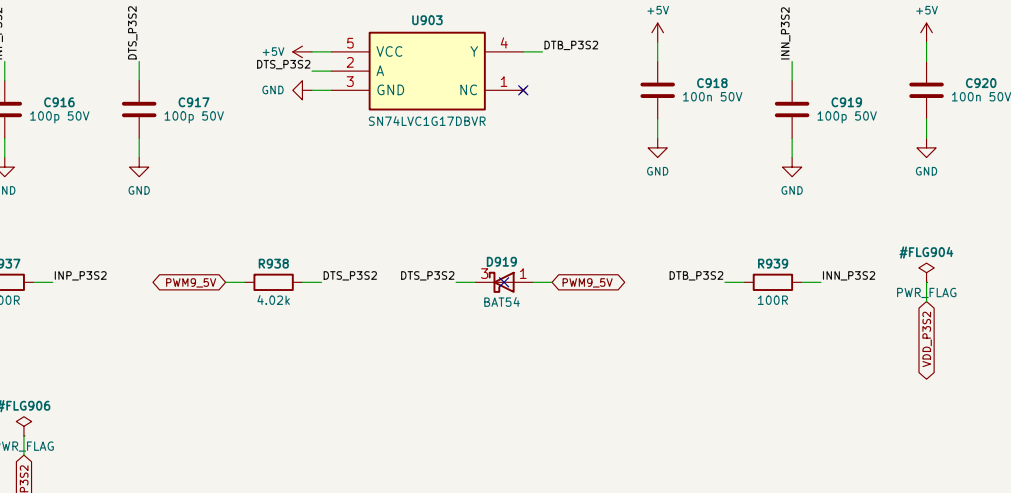
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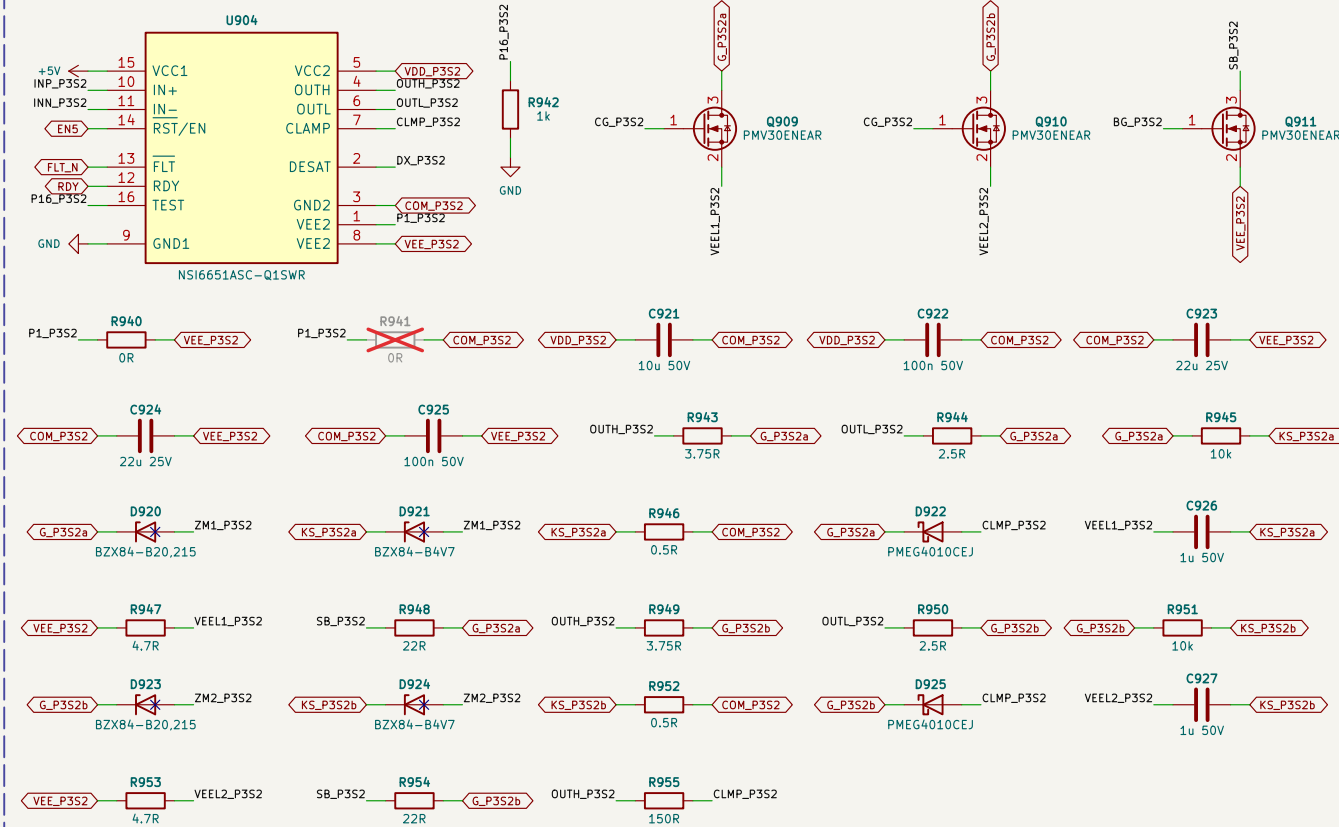
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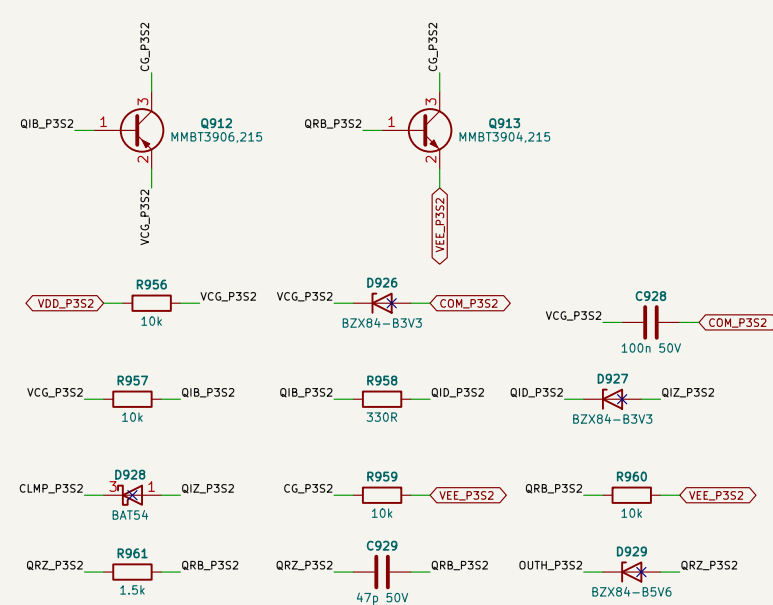
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### P3S2: DRIVER AND 2 GATES



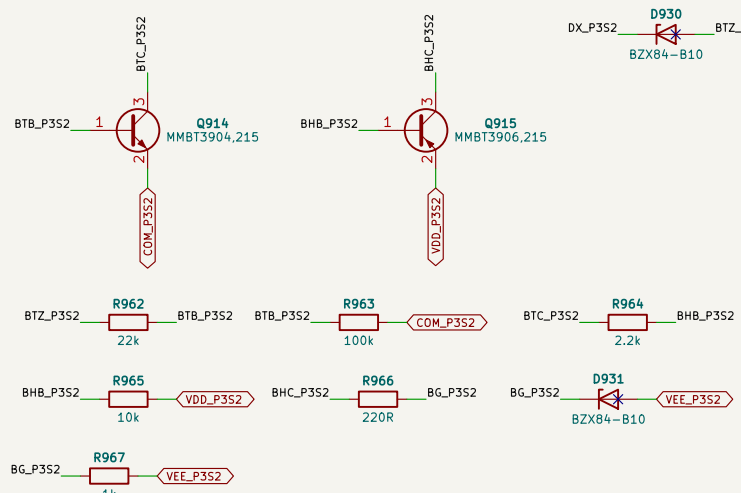
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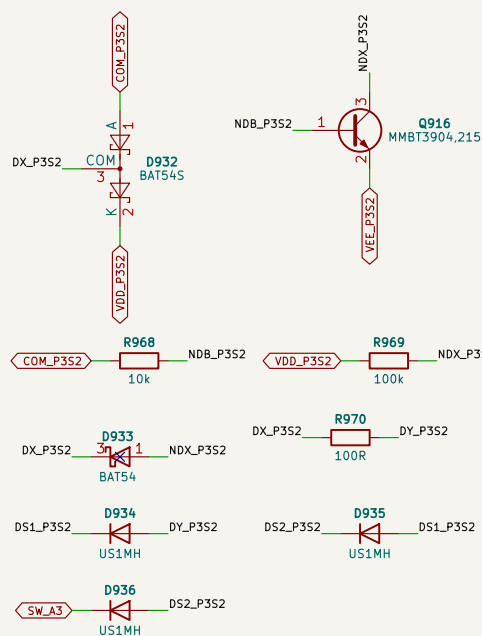
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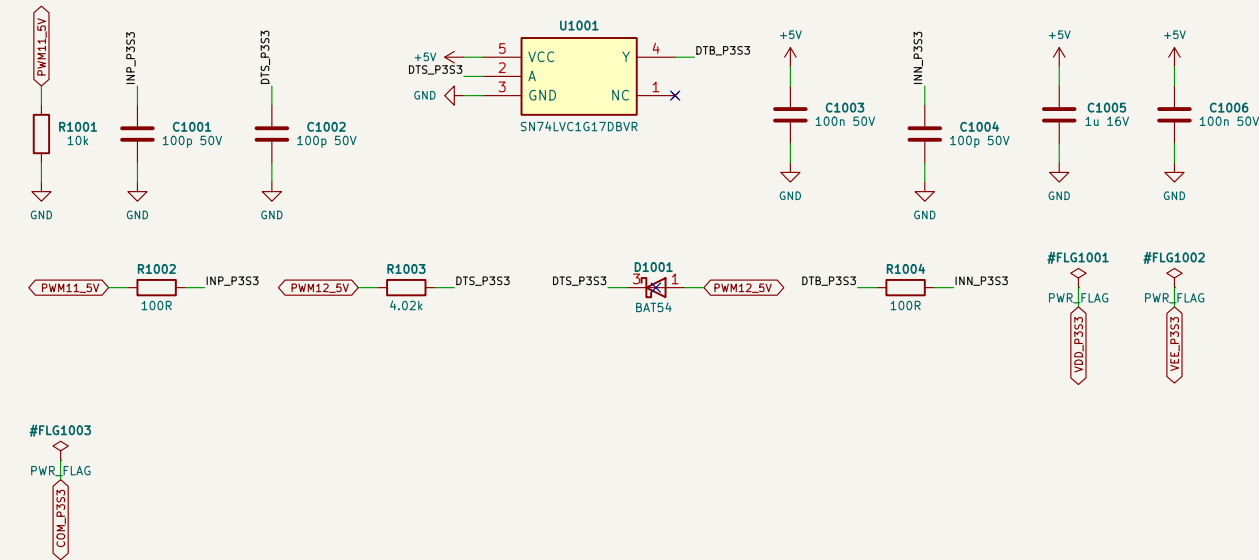


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### LEG A: LAYOUT RULES

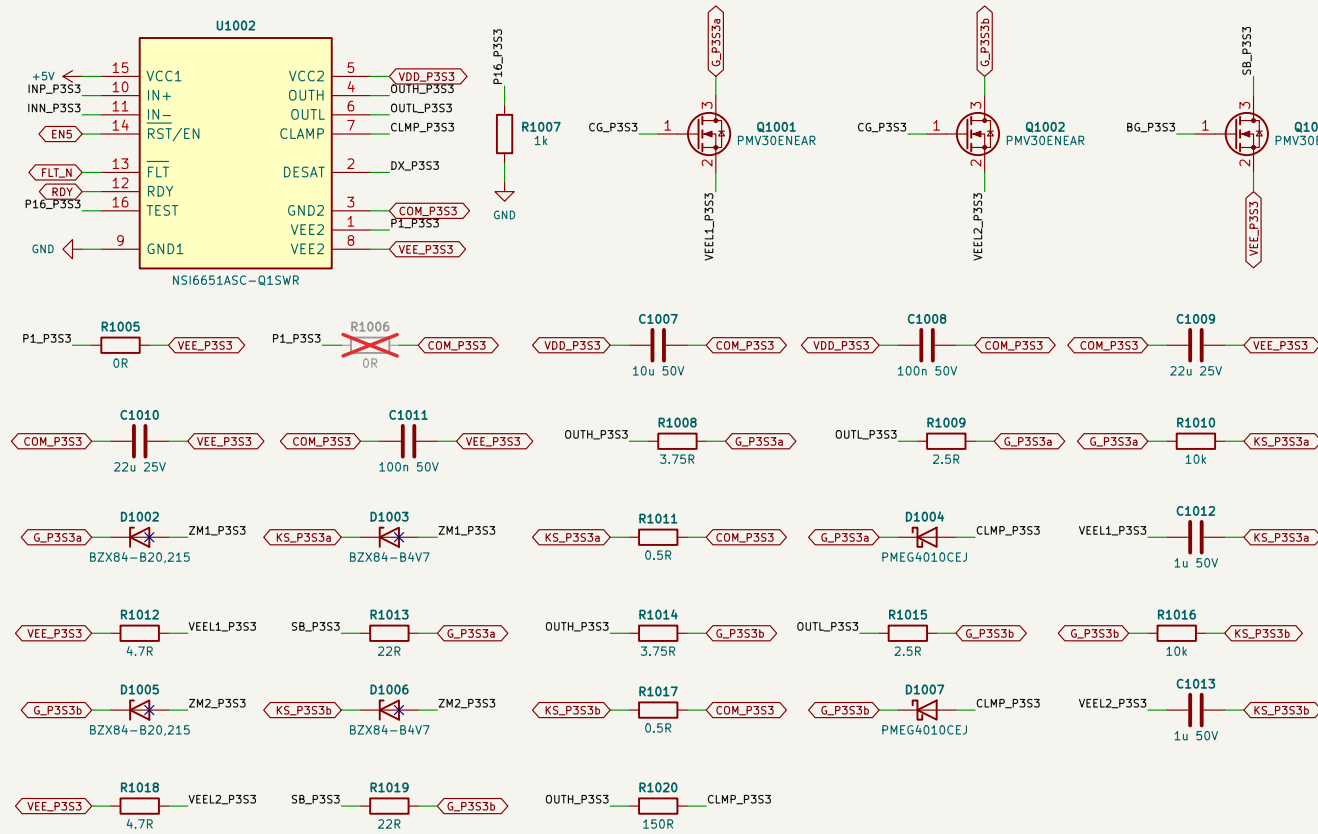
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### P3S3: LOGIC SIDE



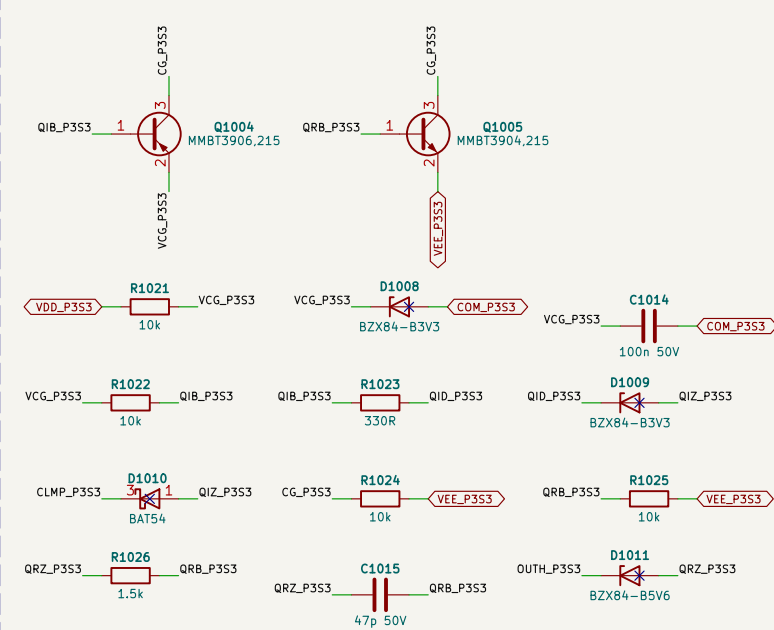
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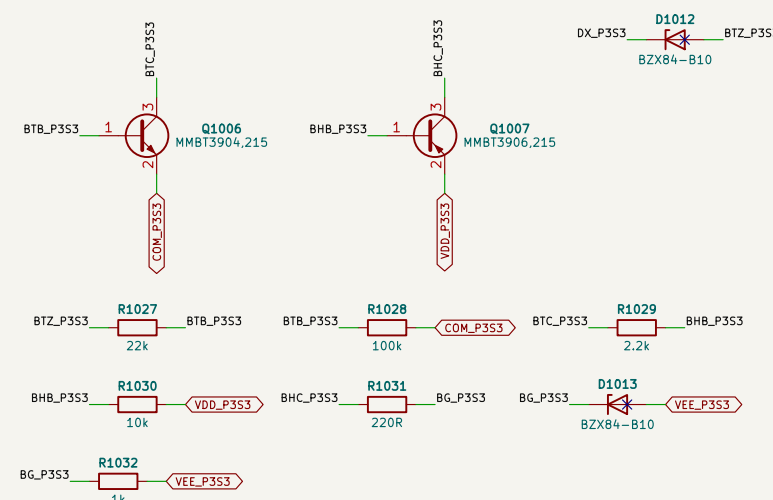
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### P3S3: MILLER CLAMP CONTROL



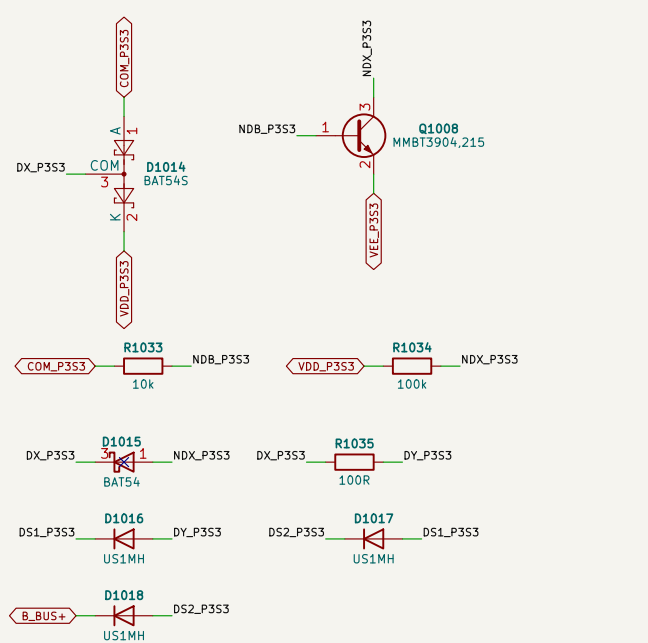
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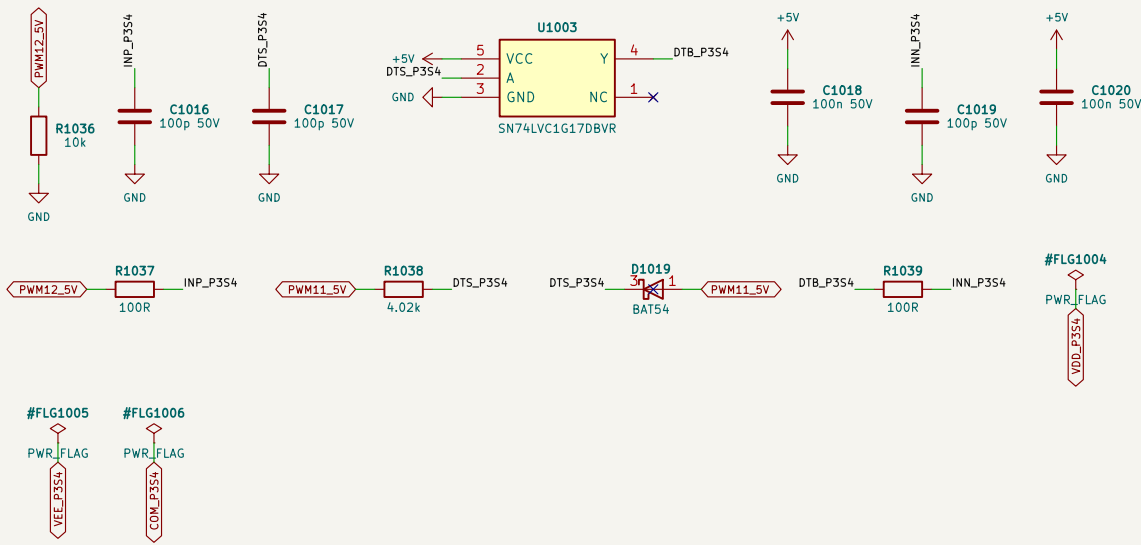
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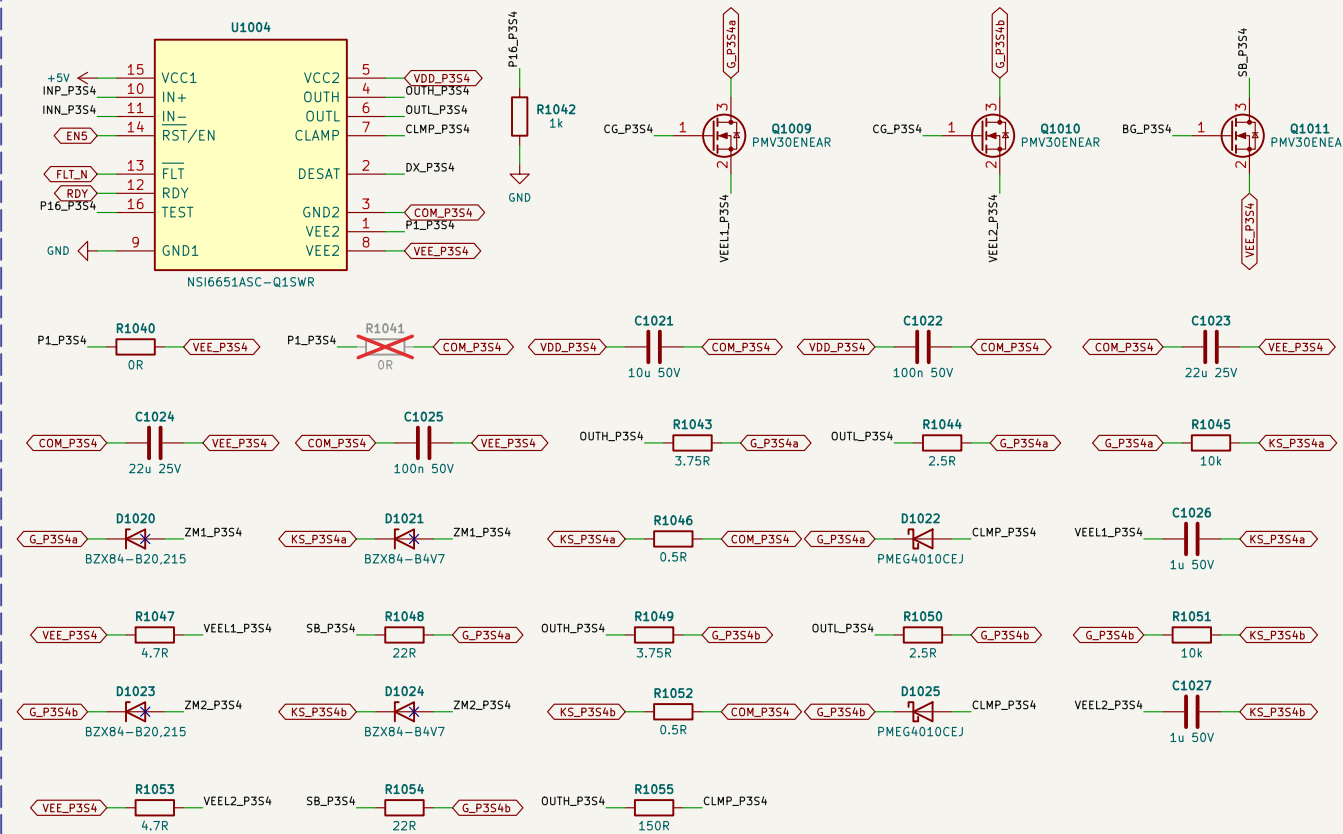
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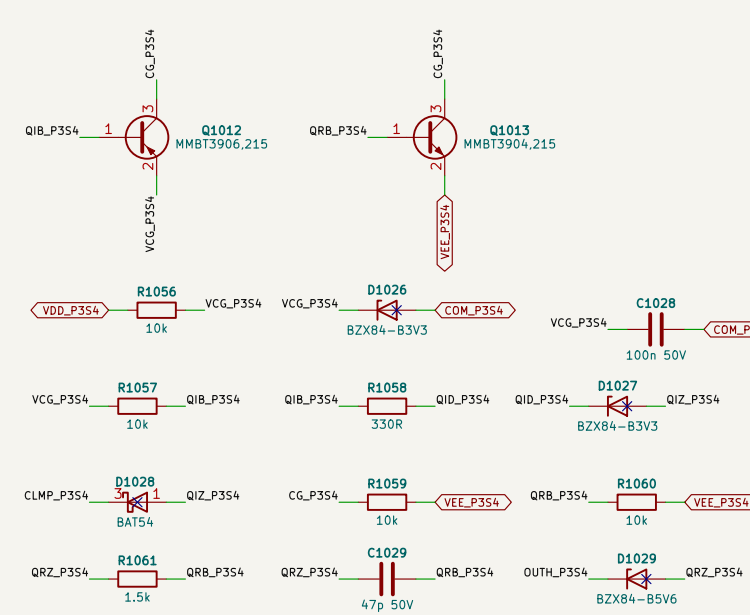
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### P3S4: DRIVER AND 2 GATES



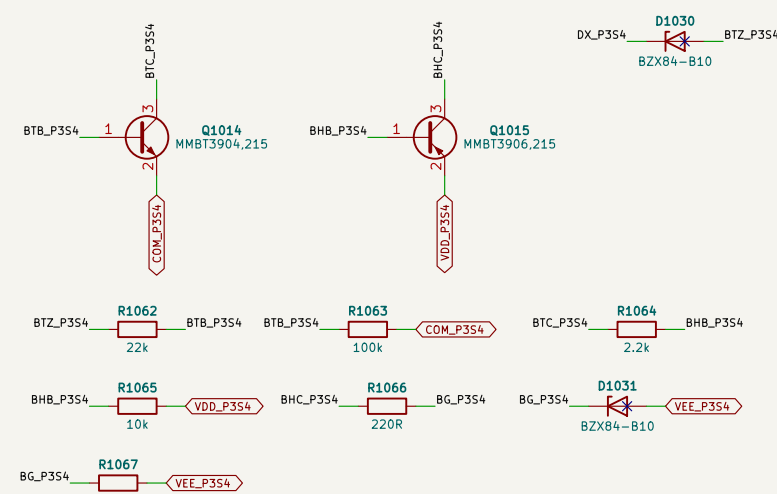
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### P3S4: MILLER CLAMP CONTROL



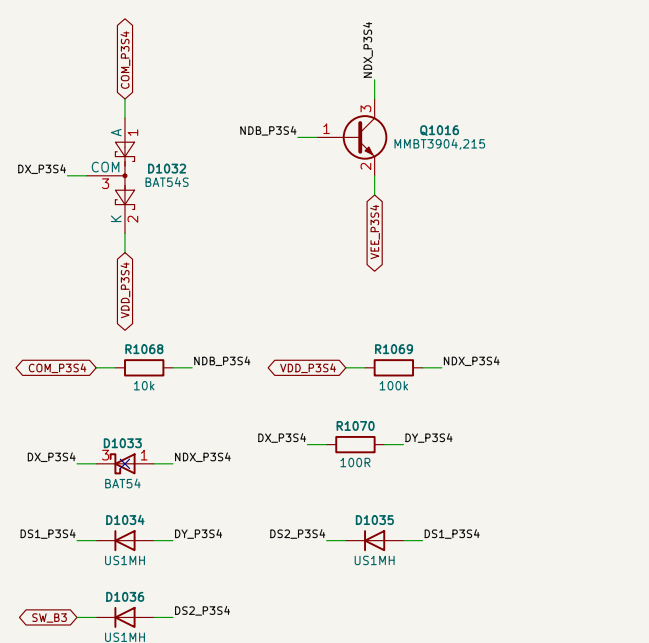
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## LEG B: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star: Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C<sub>VL</sub> at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (ngspice, gdrv): die +1.41 V vs VGS(th) min 1.94 V at 175 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

Not bench-validated. Values read from sim/out/\*.json; checks: design\_check()  
Sheet 14

Schematics authored by Chinmoy Bhuyan

Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE base

Sheet: /10 Phase 3 leg B gate drive (S3, S4)/

File: 10\_ph3\_gdb.kicad\_sch

Title: PV-PWR-4 power board - Phase 3 leg B gate drive (S3, S4)

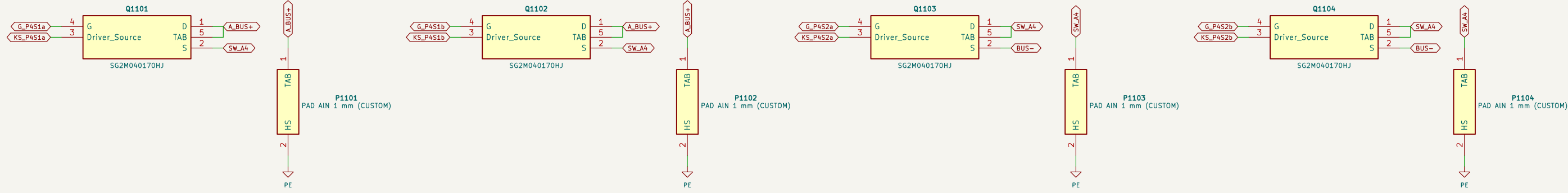
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| Size: A2 | Date: 2026-10-05 |
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Rev: A2

KiCad E.D.A. 10.0.6

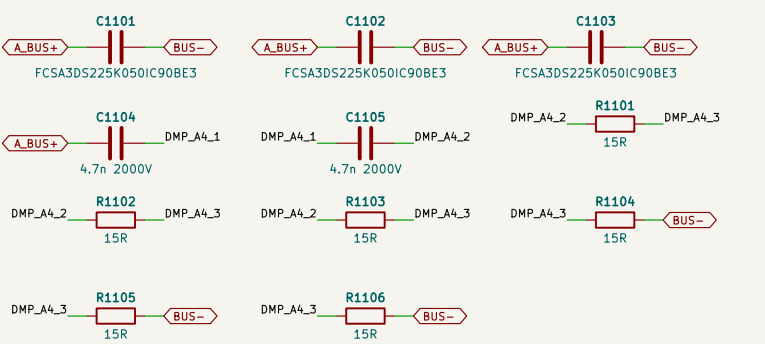
Id: 11/20

PHASE 4 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



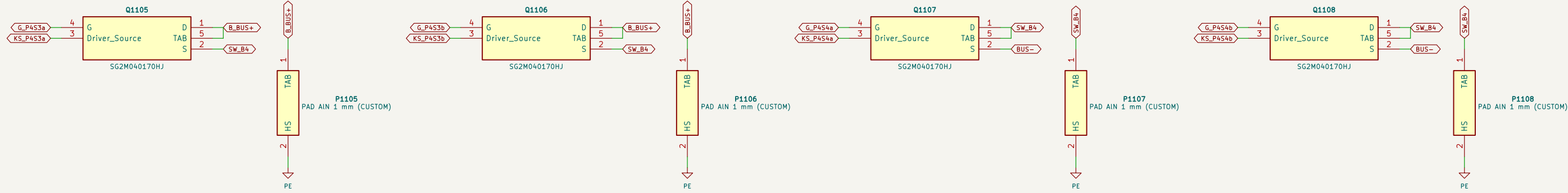
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own  
AIN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and  
pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 4 LEG A DECOUPLING + RC DAMPER



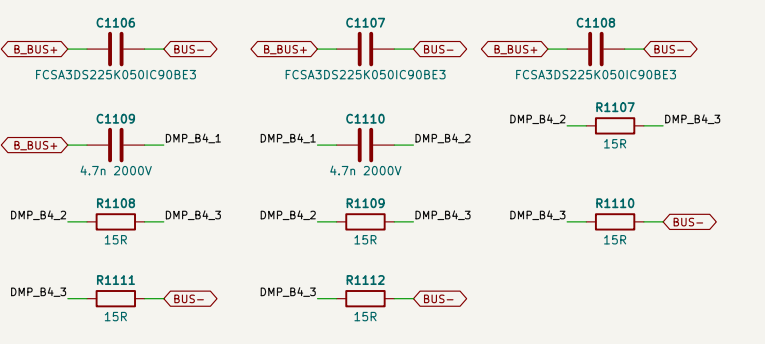
3 x FCSA3D5225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).  
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 4 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



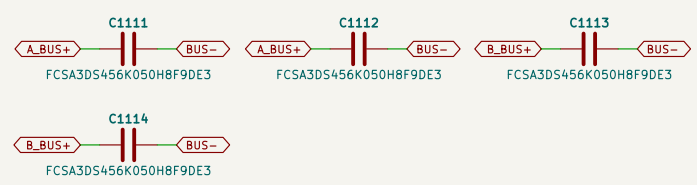
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own  
AIN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and  
pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 4 LEG B DECOUPLING + RC DAMPER



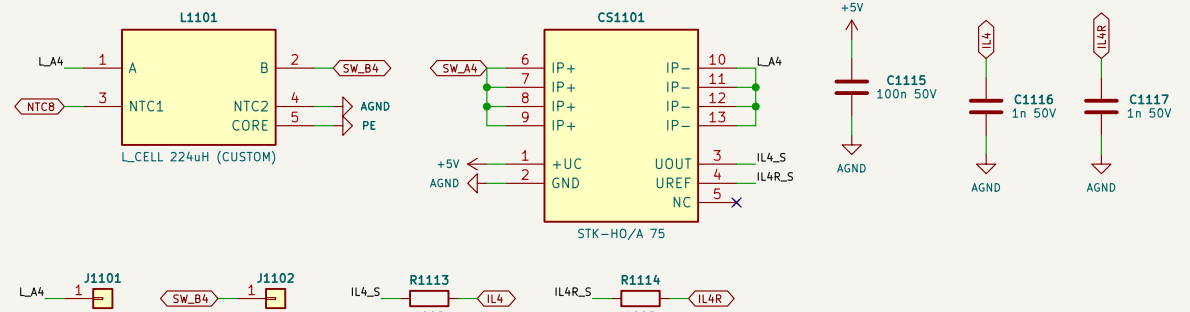
3 x FCSA3D5225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).  
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 4 SHARE OF THE PORT BANKS



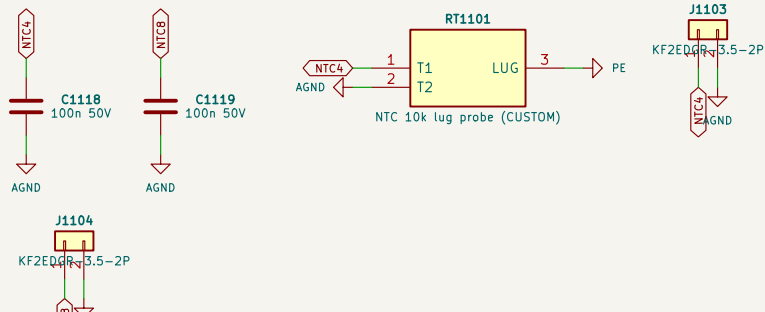
2 x FCSA3D5456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs; the banks of all phases form the  
port A / port B DC links (8 / 8 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 4 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



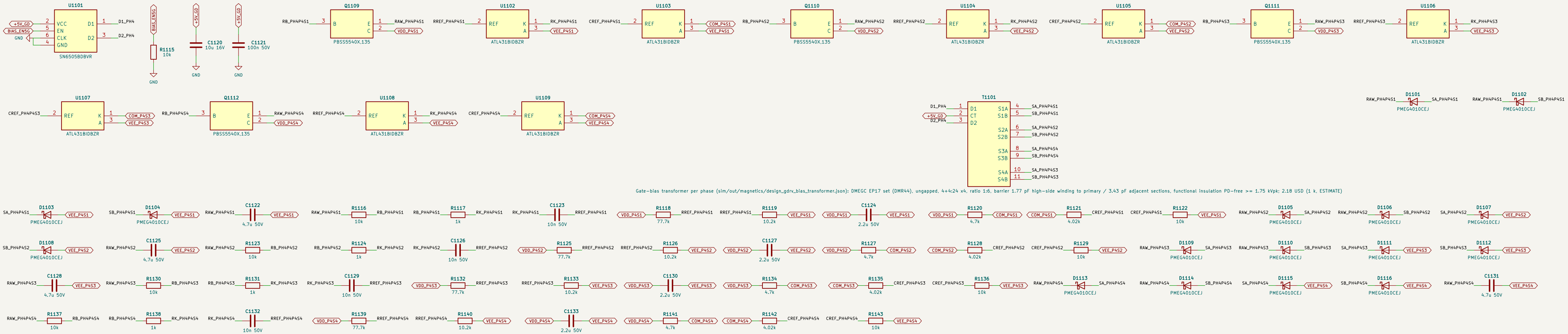
Lead A from SW\_A4 passes the STK-HQ/A 75 U-primary (pins 6-9 in, 10-13 out: + = SW\_A -> SW\_B) to stud  
L\_A4; lead B on stud SW\_B4; IL4 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V). 0.5-4.5 V over +/-187.5 A;  
100R +/- 1 mV at the source; IL4R = the sensor's own Uref (pin 4). 100R +/- 1 mV; PV-CTL references its trip  
window to it. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 4 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



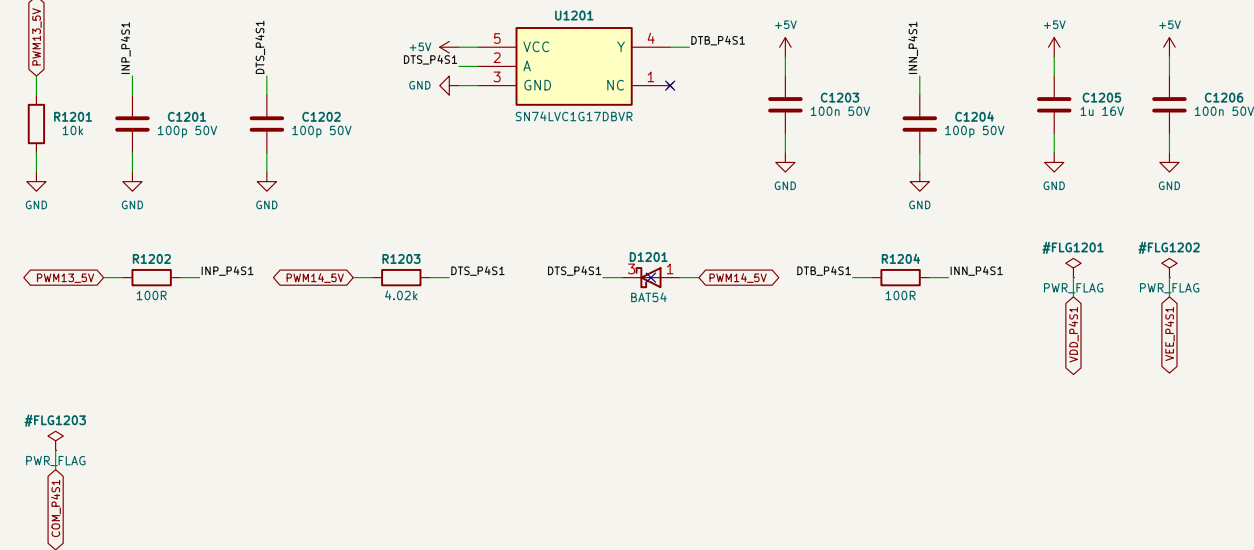
NTC4 = heatsink section of phase 4: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (8-10).  
NTC5 = inductor winding NTC (functional to the winding). 100 mV at the header against switch-node noise.

PH4: GATE BIAS (PER PHASE)



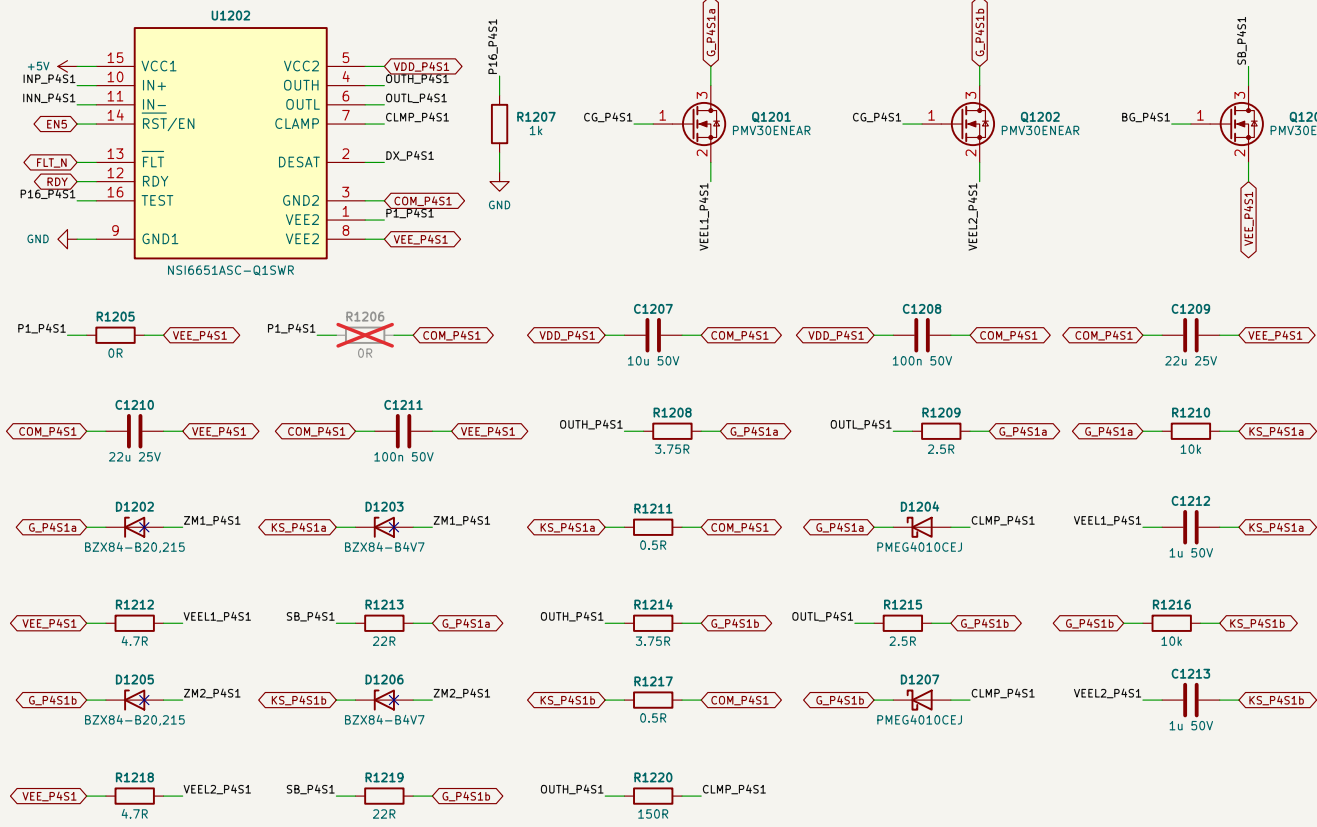
SN6505B push-pull 420 kHz from +5V\_GD, BIAS\_EN high = on; T\_BIAS4 custom, functional insulation  
(PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

### P4S1: LOGIC SIDE



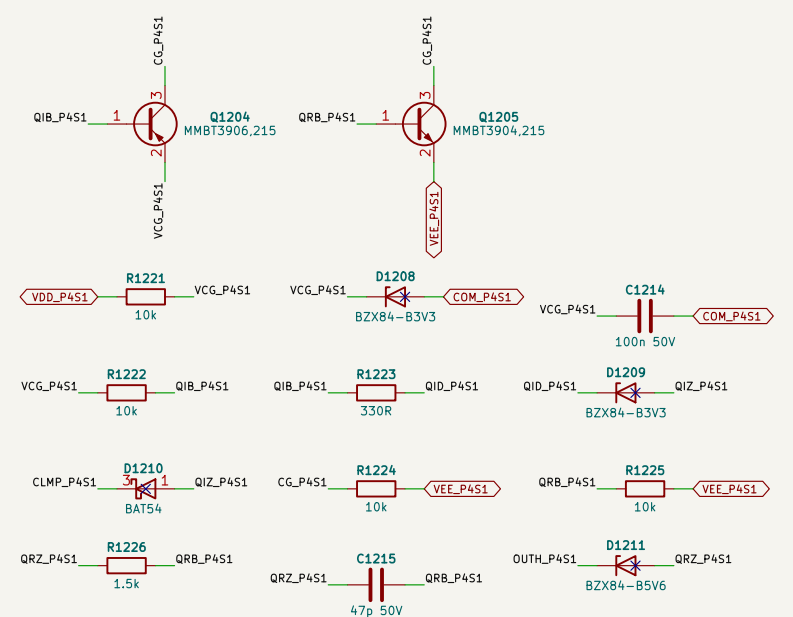
In+ = own PWM; In- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

### P4S1: DRIVER AND 2 GATES



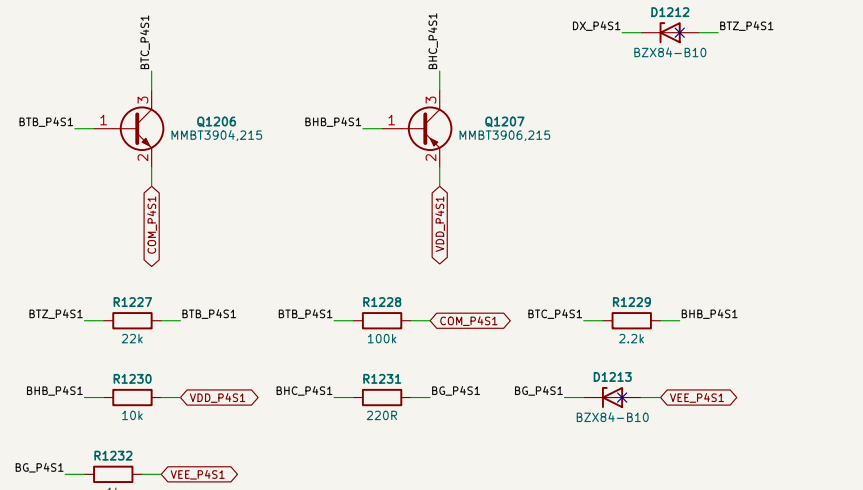
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

### P4S1: MILLER CLAMP CONTROL



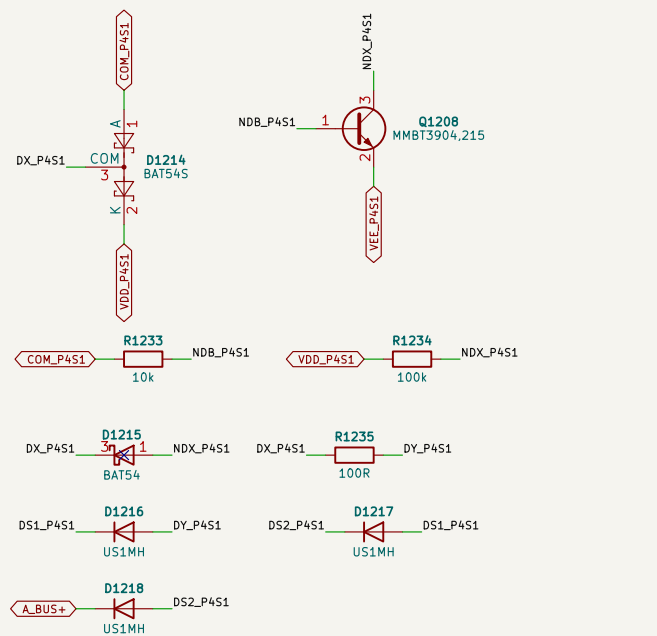
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

### P4S1: SHORT-CIRCUIT BOOSTER



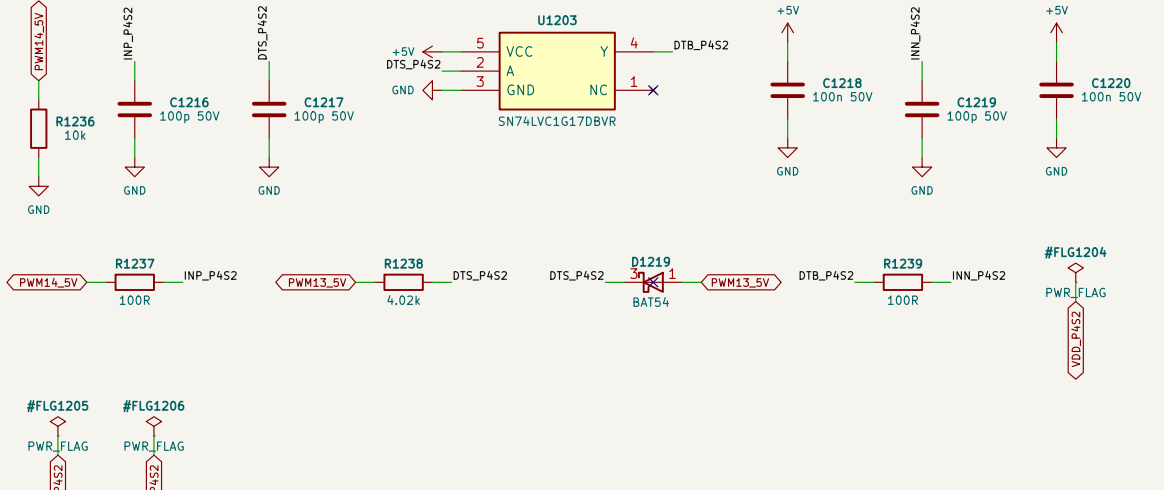
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3906 charges BG to VEE + 10 V; per device AQ340DA + 22R pull the gate to VEE; BG holds -3 us (Lk + FET gates).

### P4S1: DESAT



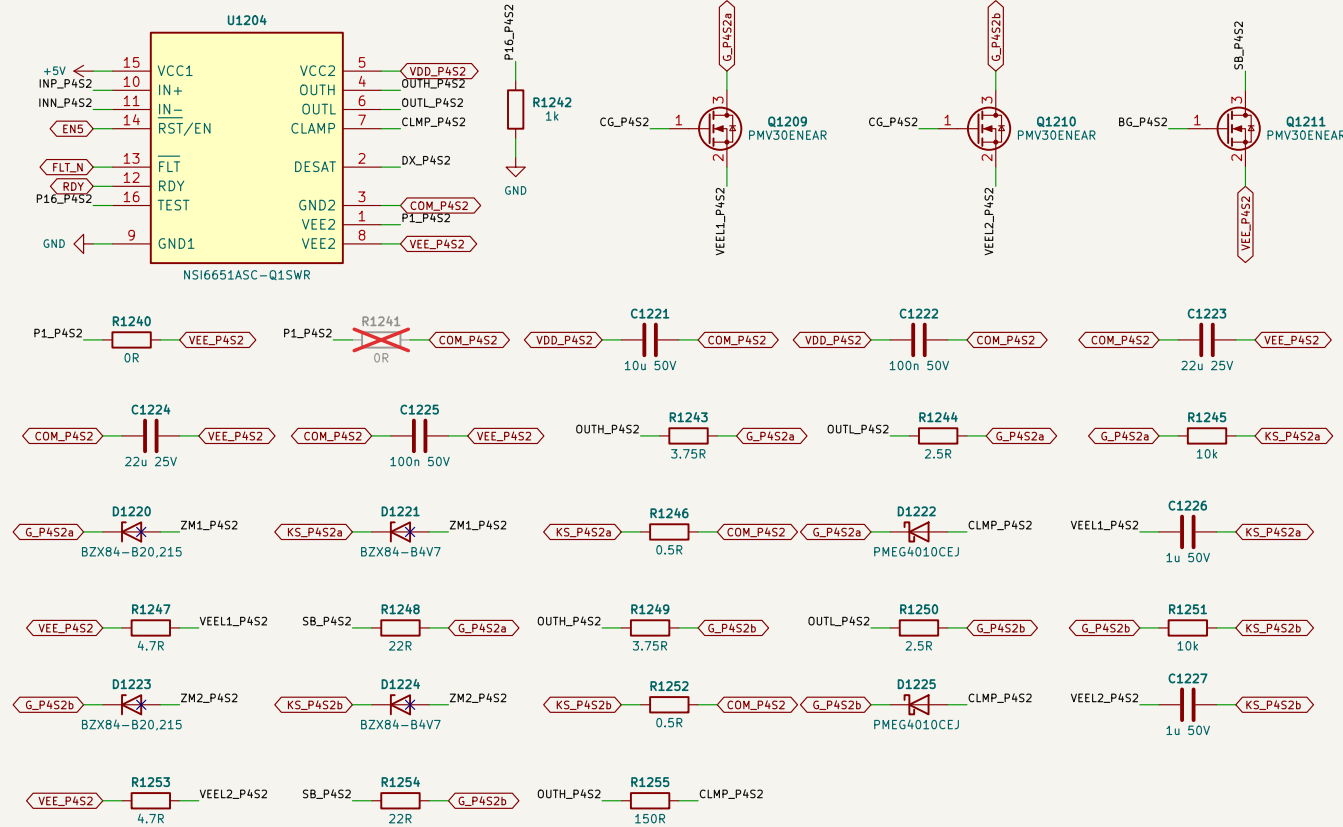
DESAT pin (internal 500 uA blanking source, trip 8.5-10.0 V) - Rk - 3 x US1M - drain. BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

### P4S2: LOGIC SIDE



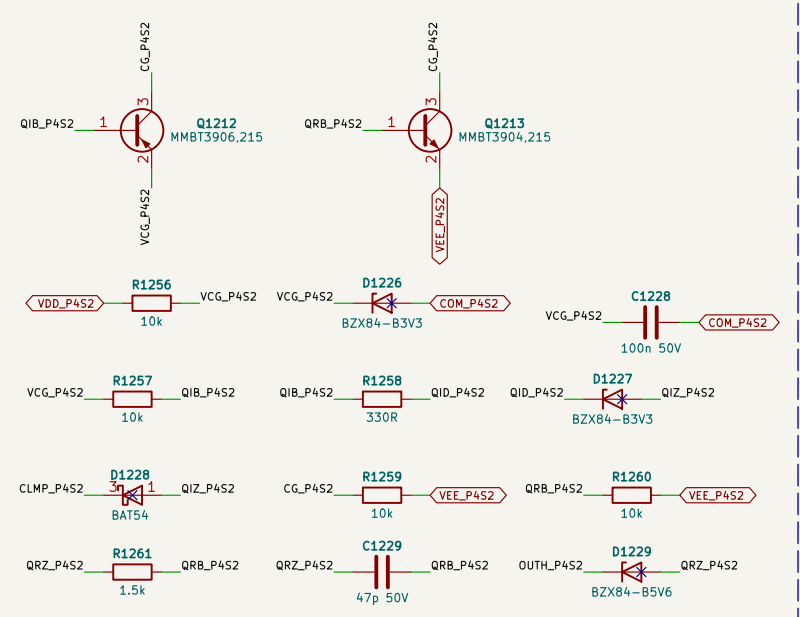
In+ = own PWM; In- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

### P4S2: DRIVER AND 2 GATES



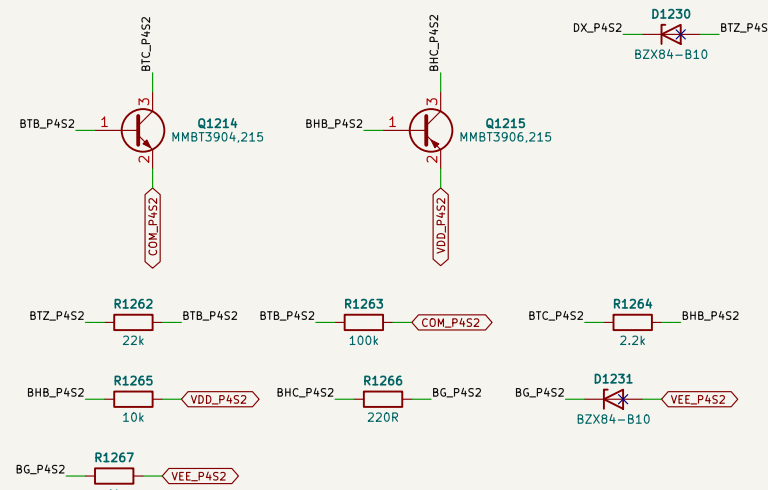
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

### P4S2: MILLER CLAMP CONTROL



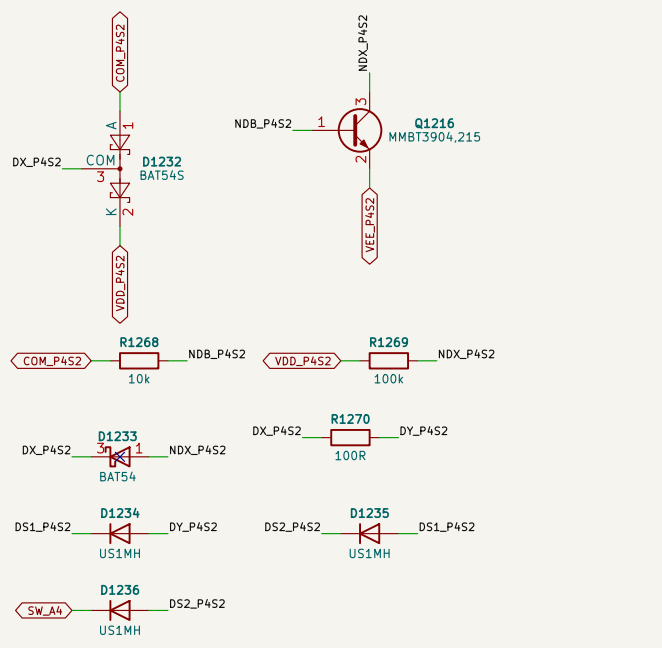
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

### P4S2: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AQ340DA + 22R pull the gate to VEE; BG holds -3 us (Lk + FET gates).

### P4S2: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5-10.0 V) - Rk - 3 x US1M - drain. BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

### LEG A: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/- 10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C<sub>VL</sub> at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string l = 2.5 mm pad gap per diode; board coated to PDI (IC-22). Miller hold (nspice, gdrv); die +1.41 V vs VGS(on) min 1.94 V at 1/5 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).



**PORT A (PV SOURCE), D-044 LEAN PORT**

Terminal side A<sub>JN</sub>- / A<sub>JN</sub>-, bank A<sub>BUS</sub>-. No fuse, no precharge; array lsc <= 1.25 x rating.  
Contactor opens <= 0.97 kA, held above (port\_spec\_lean). Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side A<sub>IN</sub>+ / A<sub>IN</sub>-, bank A<sub>BUS</sub>+. No fuse, no precharge; array I<sub>sc</sub> ≤ 1.25 x rating.  
Contactor opens ≤ 0.97 kA, held above (port\_spec lean). Installation: upstream clears 967–1250 A within 0.26 s.

TI25751 x 3 (615 VDC, thermal link + monitor lead); MOV1 +/M, MOV2 -/M, MOV3 M/PE-1, Up ref 3793 V pole-PE and pole-pole #1 in 3 kA (gALC, < 4.0 V => reinforced 6 kV / basic 4 kV, Dr=032); limax 25 kA, Branch fuses ETI CH14x51 gPV 36 A 30 kA (L/R 2 ms), Monitor loop through both fuses and all three links -> CNY65B => RBE (sum on the ID line); RBS1 = 1 at rest above 200 V = varistor network open -> CTRL inhibit/stops.

TYT25751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE.T. Up. eff 3793 V pole-P1 and pole-pole at in 3 kA (calc., <= 4.0 kV -> reinforced 6 kV / basic 4 kV, D-032); I max 25 kA. Branch fuses ETI CH14x51 gPV 36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three links -> CNY65B -> RB1 (sum on the ID line): RB1 = 1 at rest above 200 V = varistor network open -> CTRL inhibits/stops.

|                                                             |                  |           |
|-------------------------------------------------------------|------------------|-----------|
| Sheet: /14 Port A power path (pv)/                          |                  |           |
| File: 14_leanA_power.kicad_sch                              |                  |           |
| <b>Title: PV-PWR-4 power board - Port A power path (pv)</b> |                  |           |
| Size: A2                                                    | Date: 2026-10-05 | Rev: A2   |
| KiCad E.D.A. 10.0.6                                         |                  | Id: 15/26 |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| A |   |   |   |   |   |   |   |   |
| B |   |   |   |   |   |   |   |   |
| C |   |   |   |   |   |   |   |   |
| D |   |   |   |   |   |   |   |   |
| E |   |   |   |   |   |   |   |   |
| F |   |   |   |   |   |   |   |   |

|                                                                                                                                                                                                                                   |                  |                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------|
| Not bench-validated. Values read from sim/out/*.json; checks: design_check()<br>Sheet 16<br>Schematics authored by Chinmoy Bhuyan<br>Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic |                  |                |
| Sheet: /15 Port A sensing and interlocks/<br>File: 15_leanA_ctrl.kicad_sch                                                                                                                                                        |                  |                |
| Title: <b>PV-PWR-4 power board – Port A sensing and interlocks</b>                                                                                                                                                                |                  |                |
| Size: A3                                                                                                                                                                                                                          | Date: 2026-10-05 | Rev: <b>A2</b> |
| KiCad E.D.A. 10.0.6                                                                                                                                                                                                               | Id: 16/26        |                |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| A |   |   |   |   |   |   |   |   |
| B |   |   |   |   |   |   |   |   |
| C |   |   |   |   |   |   |   |   |
| D |   |   |   |   |   |   |   |   |
| E |   |   |   |   |   |   |   |   |
| F |   |   |   |   |   |   |   |   |

Not bench-validated. Values read from sim/out/\*.json; checks: design\_check()  
Sheet 17  
Schematics authored by Chinmoy Bhuyan  
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /16 Port A coil drives/  
File: 16\_leanA\_drive.kicad\_sch

Title: **PV-PWR-4 power board – Port A coil drives**

Size: A3

Date: 2026-10-05

Rev: **A2**

KiCad E.D.A. 10.0.6

Id: 17/26

Terminal side B<sub>LN</sub>+ / B<sub>LN</sub>-, bank B<sub>BUS</sub>+, HIPE501 250 A wR in both poles (breaks 1.25-50 kA only).  
Contactor opens <= 0.97 kA, held above (port\_spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side B<sub>LN</sub>+ / B<sub>LN</sub>-, bank B<sub>BUS</sub>+, HIPE501 250 A wR in both poles (breaks 1.25-50 kA only).  
Contactor opens <= 0.97 kA, held above (port\_spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

TV125751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE-T,  $U_{eff}$  3793 V pole-PE and pole-pole at 3 kA (calc. = 4.0 kV) => reinforced 6 kV / basic 4 kV, D=032; Imax 25 kA, Branch fuses ITI CHM45 gGpV 36 A (30 kA / I<sub>2t</sub> 2 ms). Monitor loop through both fuses and all three links -> CNY65B -> RB1 (sum on the ID line): RB1 = 1 at rest above 200 V = varistor network open -> CTRL inhibits/stops.

HPE501/000B100-250 in both poles: breaks 1.25-50 kA only (partial range)  
the part OC trip and the contactor keep it out of 250 A-1.25 kA.

HPE501/000B100-250 in both poles: breaks 1.25-50 kA only (partial range)  
the part OC trip and the contactor keep it out of 250 A-1.25 kA.

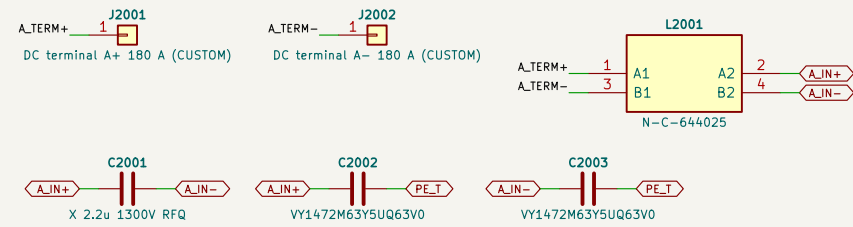
|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| A |   |   |   |   |   |   |   |   |
| B |   |   |   |   |   |   |   |   |
| C |   |   |   |   |   |   |   |   |
| D |   |   |   |   |   |   |   |   |
| E |   |   |   |   |   |   |   |   |
| F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

|                                                                                                                                                                                                                                   |                  |         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------|
| Not bench-validated. Values read from sim/out/*.json; checks: design_check()<br>Sheet 19<br>Schematics authored by Chinmoy Bhuyan<br>Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic |                  |         |
| Sheet: /17 Port B sensing and interlocks/<br>File: 17_leanB_ctrl.kicad_sch                                                                                                                                                        |                  |         |
| Title: <b>PV-PWR-4 power board – Port B sensing and interlocks</b>                                                                                                                                                                |                  |         |
| Size: A3                                                                                                                                                                                                                          | Date: 2026-10-05 | Rev: A2 |
| KiCad E.D.A. 10.0.6                                                                                                                                                                                                               | Id: 19/26        |         |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| A |   |   |   |   |   |   |   |   |
| B |   |   |   |   |   |   |   |   |
| C |   |   |   |   |   |   |   |   |
| D |   |   |   |   |   |   |   |   |
| E |   |   |   |   |   |   |   |   |
| F |   |   |   |   |   |   |   |   |

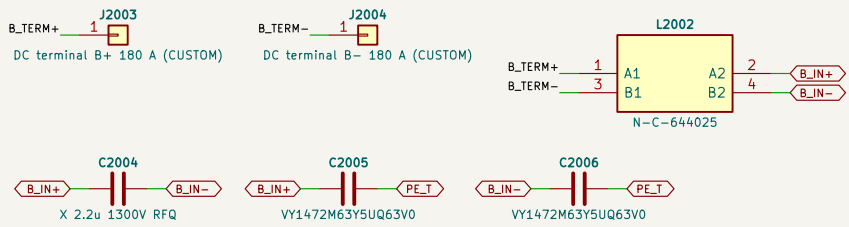
|                                                                                                                                                                                                                                   |                  |                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------|
| Not bench-validated. Values read from sim/out/*.json; checks: design_check()<br>Sheet 20<br>Schematics authored by Chinmoy Bhuyan<br>Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic |                  |                |
| Sheet: /18 Port B coil drives/<br>File: 18_leanB_drive.kicad_sch                                                                                                                                                                  |                  |                |
| Title: <b>PV-PWR-4 power board – Port B coil drives</b>                                                                                                                                                                           |                  |                |
| Size: A3                                                                                                                                                                                                                          | Date: 2026-10-05 | Rev: <b>A2</b> |
| KiCad E.D.A. 10.0.6                                                                                                                                                                                                               | Id: 20/26        |                |

PORT A TERMINALS AND EMI FILTER



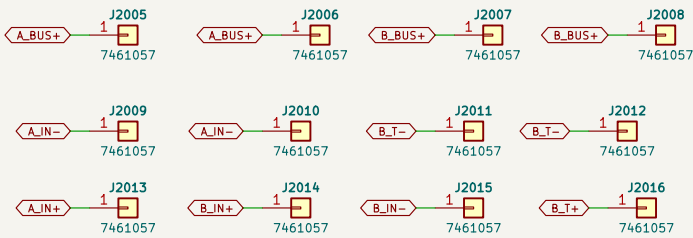
PV: chassis terminals A\_TERM+/- (180 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1). X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE\_T (removable PE link for the hipot). Board taps A\_IN+ by short links.

PORT B TERMINALS AND EMI FILTER



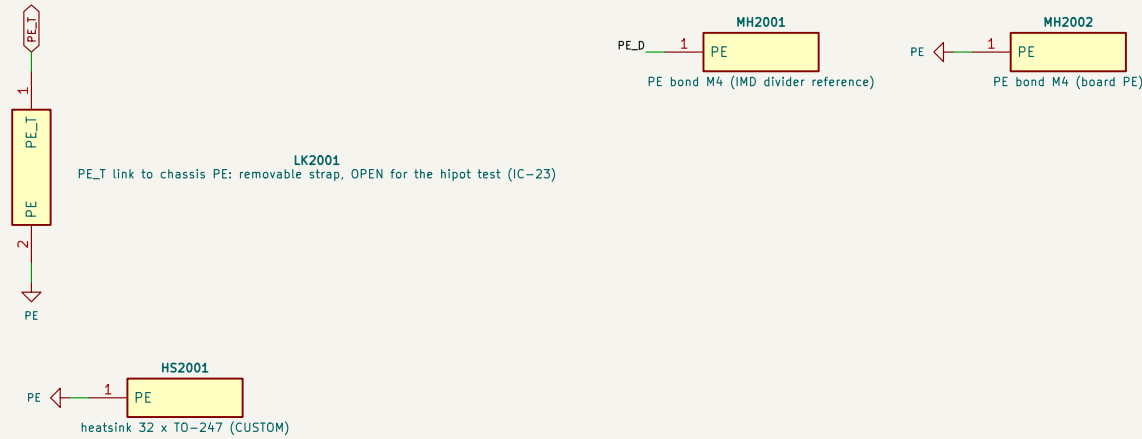
battery: chassis terminals B\_TERM+/- (180 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1). X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE\_T (removable PE link for the hipot). Board taps B\_IN+ / B\_IN- / B\_T+ by short links.

BOARD ENTRIES (WURTH REDCUBE PRESS-FIT M3, 100 A EACH)



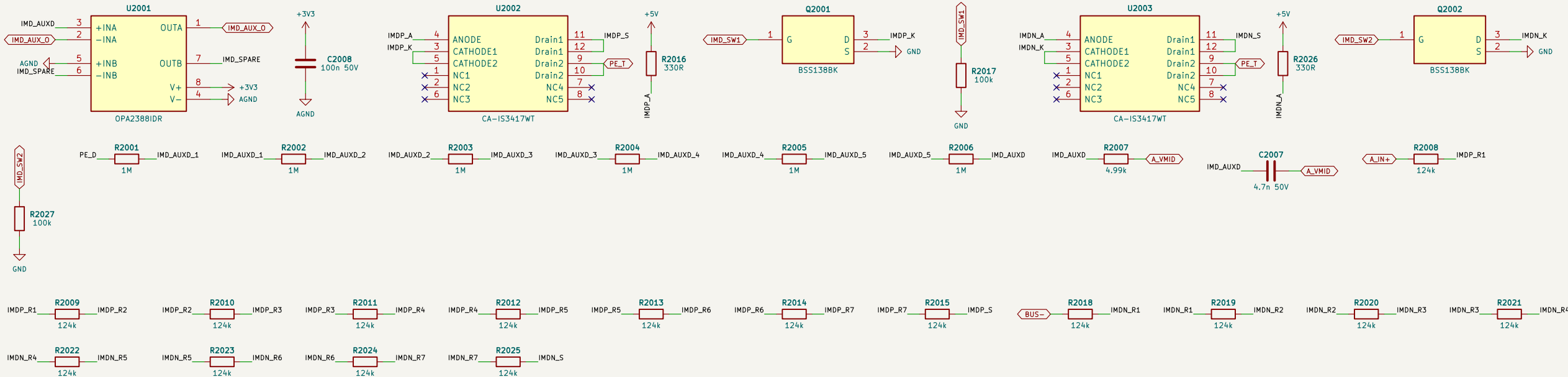
Bus entries from the contactors / fuses: A-BUS+, B-BUS+, A\_IN- and B\_T- (into the shunts), two studs each; taps A\_IN+, B\_IN+, B\_IN-, B\_T+ (varistors, dividers, IMD, precharge, aux input), one each.

PE: LINKS, HEATSINK



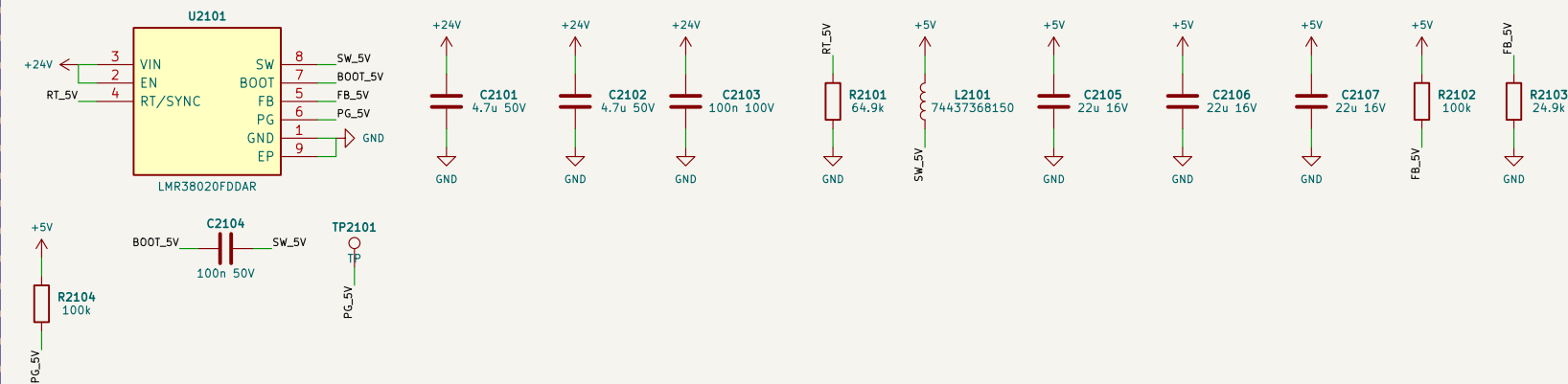
PE\_T (Y capacitors, MOV3, IMD strings) reaches PE through a removable strap (open for the hipot, IC-23); PE\_D = IMD divider bond; the heatsink is earthed at two points.

INSULATION MONITOR (PORT.LEAN\_IMD)



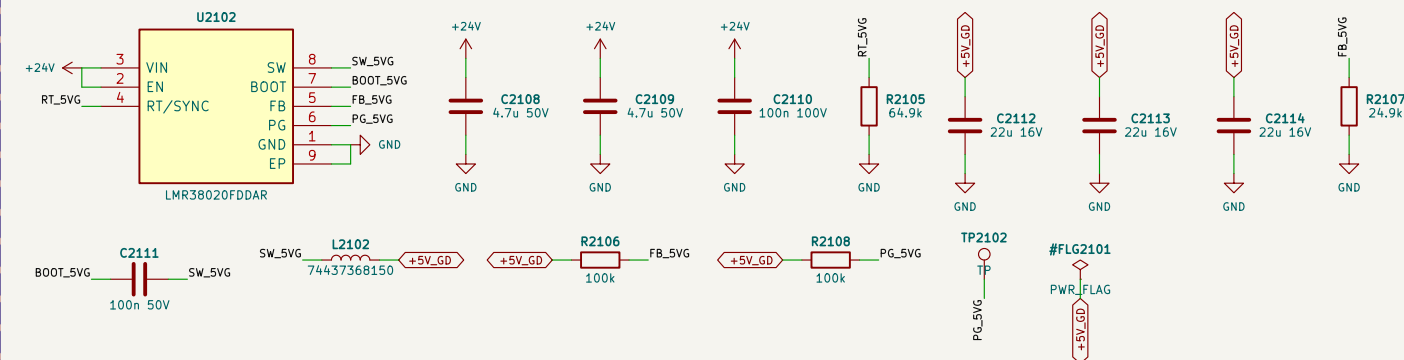
IMD\_SW1 = string A+ -> PE, IMD\_SW2 = string PE -> BUS- (high = connected); VPE = PE divider against BUS- (VMID of port A).

24 V  $\rightarrow$  +5V (LMR38020, 400 KHZ, 2 A)



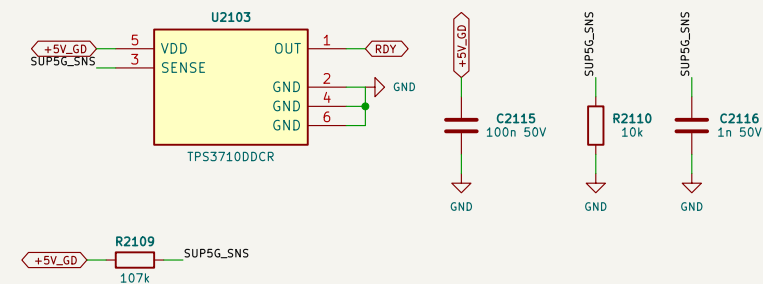
NS16651 inputs, AHCT, sensors, references, PC +5V, gate bias of phases 1-2. ctr\_lc2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design\_check (<= 80 % of 2 A).

24 V -> +5V\_GD (LMR38020, 400 KHZ, 2 A)



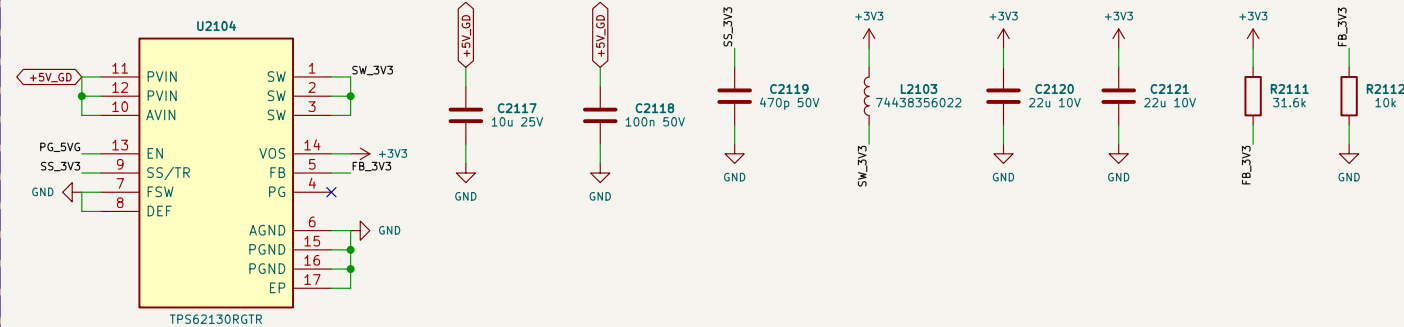
gate bias of phases 3-4. ctrlC2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design\_check (<= 80 % of 2 A).

## +5V\_GD SUPERVISOR ON RDY



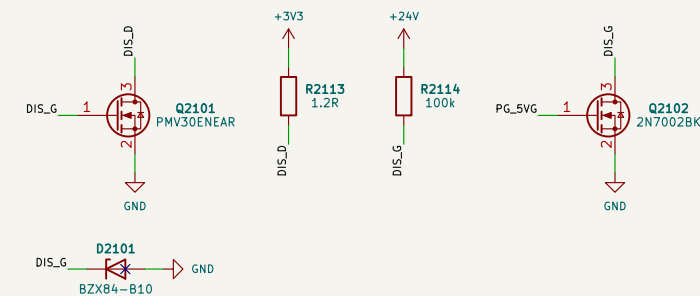
TPS3710 107k/10k as on +5V: RDY low while the bias rail of phases 3-4 is below 4.52-4.69 V.

**+5V\_GD -> +3V3 (TPS62130, 2.5 MHZ) WITH A CONTROLLED RAMP**



EN = PG\_5V0, FB 31.6k / 10k 0.1 % -> 3.240-3.433 V ( $\geq 3.20$  V for the control board's reference), SS/TR 470p COG; ramp-up 12-16 mV/us (F28003x VDDIO 8-100 mV/us). Ports' logic, RDY / MOV\_OK pull-ups, PC +3V3 (control  $\leq 157$  mA).

### +3V3 DISCHARGE (RAMP-DOWN WHEN THE RAILS FAIL)



PG\_5VG low (+5V\_GD < 90-94 %, i.e. the live 24 V is gone) disables the TPS62130 and lets the 100k from +24V (10 V Zener) turn on PMV30ENEA + 1.2R: +3V3 falls at  $\geq 20$  mV/us through the BOR region (F28003x SRVDDIO-DN).

**SELV 24 V TO THE CONTROL BOARD (J2)**



AUX-T1 reinforced winding SELV<sub>24V</sub> / SELV<sub>0V</sub>, cross-regulated 21.4–30.1 V (aux75\_spec); full fan speed needs 24.5 V. Full fan speed is only guaranteed while the converter is switching. Fan buck (12–36 V in) + SELV logic on the control board; >= 10 mm from live copper; nothing else of the SELV domain is on this board.

Not bench-validated. Values read from sim/out/\*.json; checks: design\_check()

Sheet 22

Schematics authored by Chinmoy Bhuyan

Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /19 Live 5 V / 3.3 V, SELV connector/

File: 19\_supplies.kicad\_sch

**Title: PV-PWR-4 power board – Live 5 V / 3.3 V, SELV connector**

Size: A3

Date: 2026-10-05

Rev: A2

KiCad E.D.A. 10.0.6

Id: 22/26

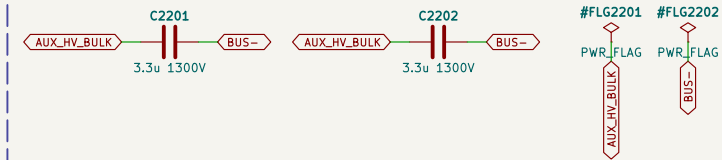


2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.64 W each at the highest input current (0.48 A, 250 V)  
Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current  $\leq 25$  A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting



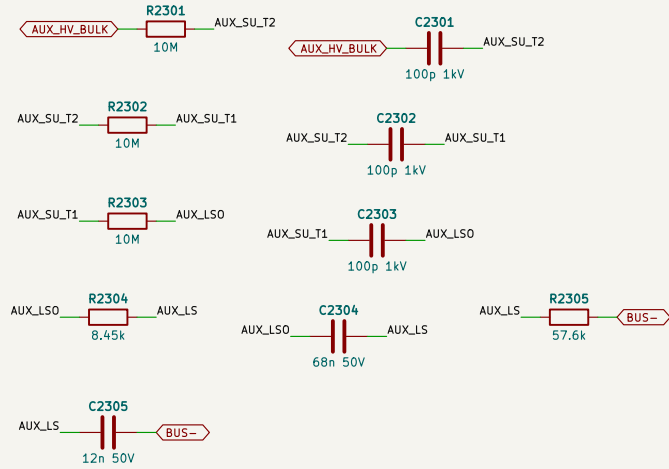
2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.64 W each at the highest input current (0.48 A, 250 V)  
Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current  $\leq 25$  A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

**SURGE ABSORBER / INPUT FILTER**



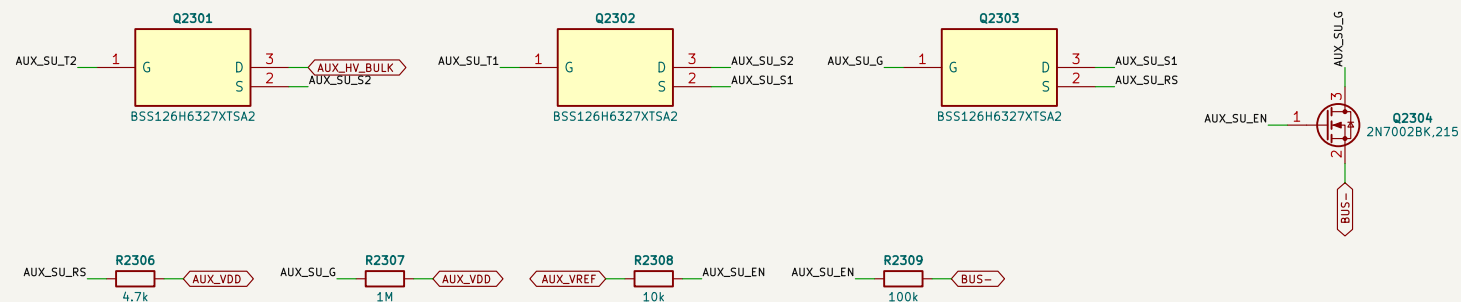
Film: (30 W case, unchanged) 2 x film = RC surge absorber with 44R: 4 kV / 20 us residual → 1199 V (UR 1300 V); filter corner 546 Hz.

COMPENSATED TAP STRING



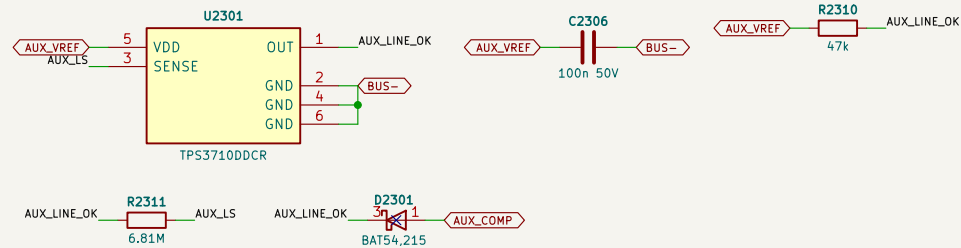
string tap, brown-in 206–215 V, brown-out 179–192 V  
brown-in hysteresis (same solve); OV lockout 1107–1162 V

DEPLETION CASCODE CURRENT SOURCE



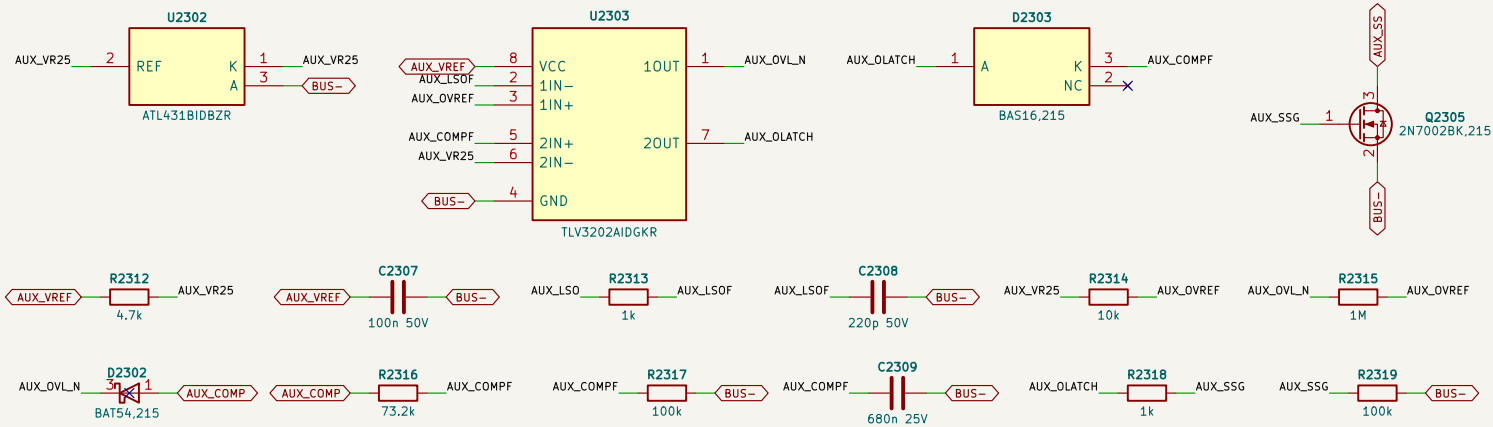
VREF high -> start-up off (as AUX-HV).

LINE BROWN-IN



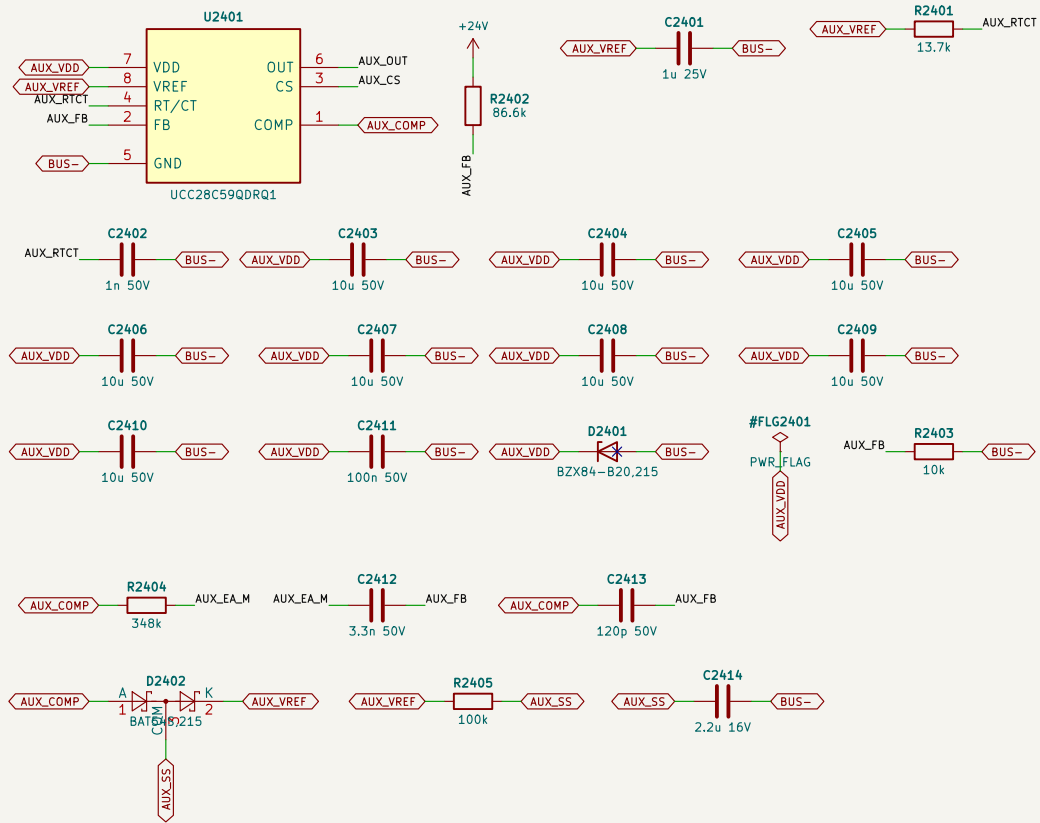
LINE\_OK low -> BAT54 holds COMP below the 1.15 V offset.

INPUT OV LOCKOUT AND OVERLOAD TIMER



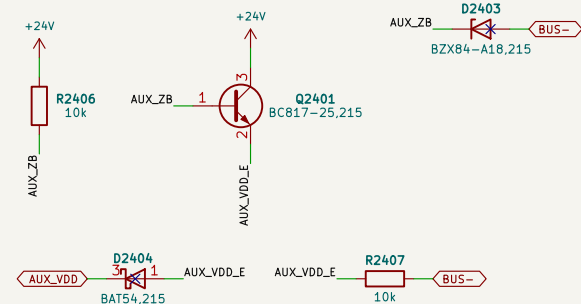
Ch 1: OV lockout -> COMP low. Ch 2: timer divider: trips above 113 / 175 / 278 W (no trip at the 112.2 W 1 s peak); timer 39–100 ms -> latch -> SS crowbar -> hiccup.

PWM CONTROLLER AND ERROR AMPLIFIER



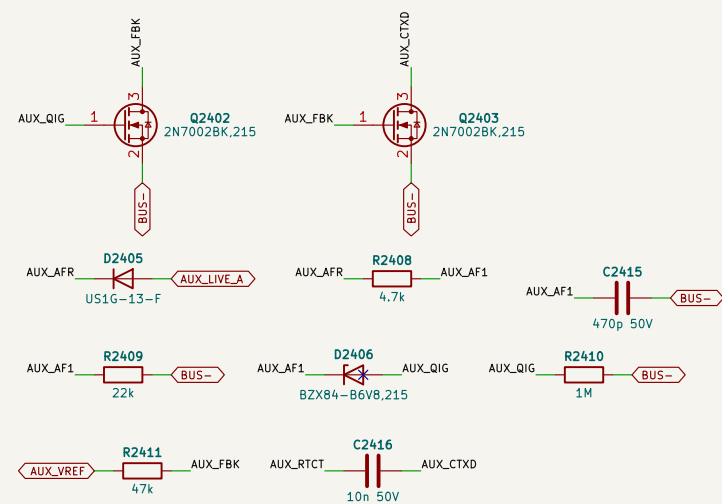
live 24 V into FB: 23.62 / 24.15 / 24.68 V (VFB 2.45–2.55 V, 0.1 %)  
type II: mid-band gain 4.0; crossover 259–727 Hz, phase margin >= 60 deg type II zero 139 Hz type II pole 3811 Hz

VDD FOLLOWER FROM THE LIVE 24 V



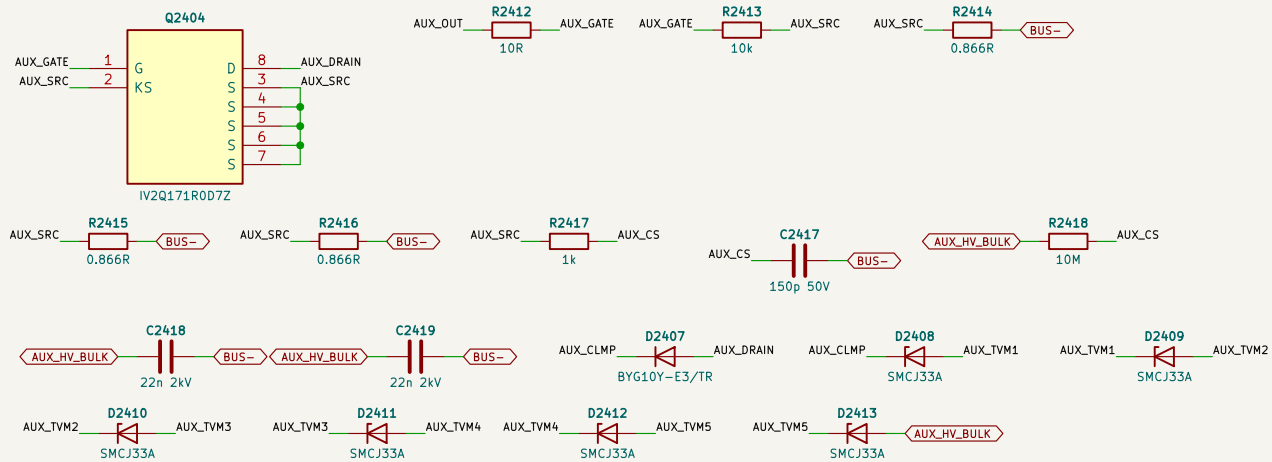
(30 W case, unchanged) follower Zener bias: VDD = gate drive 15.46–18.11 V (BZX84–A18, –40..85 C)

FREQUENCY FOLDBACK ON THE LIVE WINDING PLATEAU (AX–03)



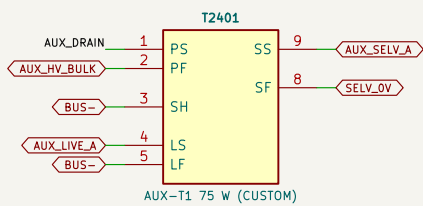
(30 W case, unchanged) foldback: switched onto RT/CT below 7.7–10.1 V output -> fsw 5.9 kHz (5.3–6.8)

SIC SWITCH, CURRENT SENSE, LINE FEED-FORWARD, CLAMP



3 in parallel = 0.288667R (E96); lowest limit 3.04 A delivers 112.8 W (>= the 112.2 W 1 s peak) at every input; highest 4.07 A with the line feed-forward (4.46 A without)  
line feed-forward HV\_BULK -> CS (CRHV2512); CS offset 101 mV at 1000 V cancels most of the 155–250 ns CS-delay overshoot, so the highest current limit no longer rises with the input  
Clamp: BYG10Y + 6 x SMCJ33A; V\_DS 1318 / 1422 V at 1000 / 1100 V.

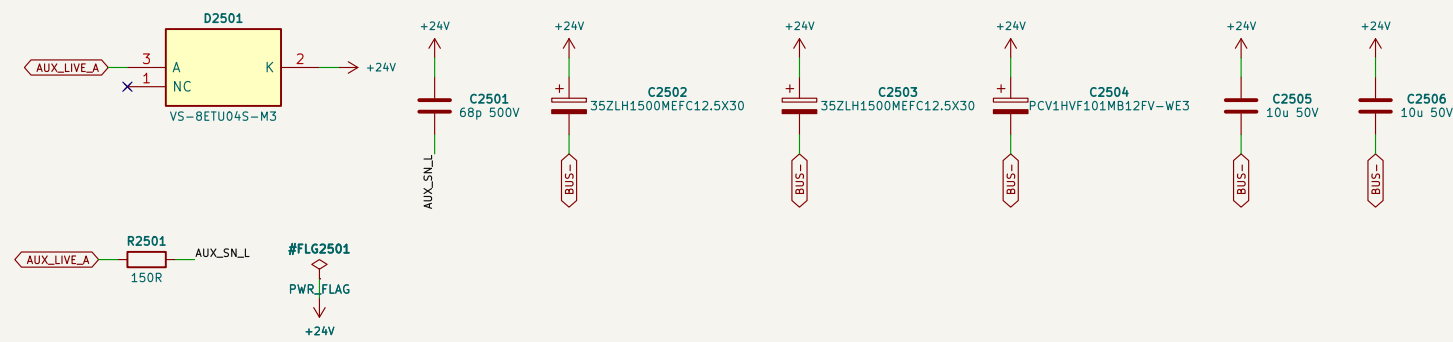
AUX-T1 (CUSTOM – THE MODULE'S REINFORCED SUPPLY BARRIER B1)



Lp 0.485 mH +/-7 %; shield and live winding on BUS-; SELV winding reinforced (PD <= 10 pC routine test).  
Electrical requirement: aux75\_spec.json transformer\_requirement.

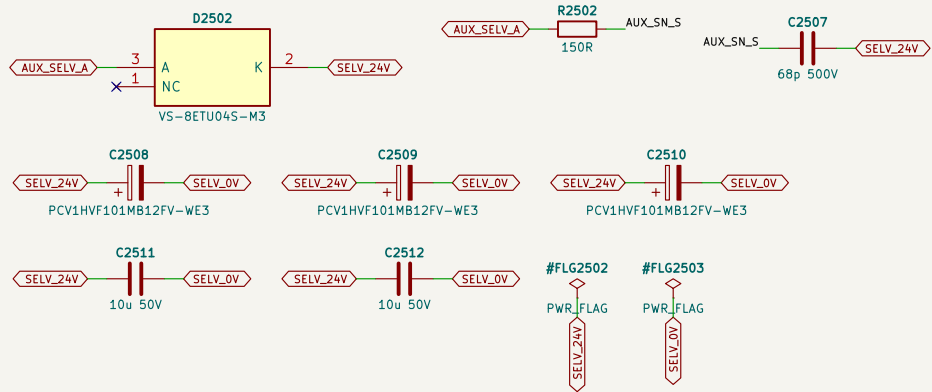
|                                                                                                                                                                                                                                   |                  |           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------|
| Not bench-validated. Values read from sim/out/*.json; checks: design_check()<br>Sheet 25<br>Schematics authored by Chinmoy Bhuyan<br>Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic |                  |           |
| Sheet: /22 AUX75 PWM, SiC, clamp, AUX-T1/<br>File: 22_aux75_power.kicad_sch                                                                                                                                                       |                  |           |
| <b>Title: PV-PWR-4 power board – AUX75 PWM, SiC, clamp, AUX-T1</b>                                                                                                                                                                |                  |           |
| Size: A3                                                                                                                                                                                                                          | Date: 2026-10-05 | Rev: A2   |
| KiCad E.D.A. 10.0.6                                                                                                                                                                                                               |                  | Id: 25/26 |

LIVE 24 V RECTIFIER, HOLD-UP



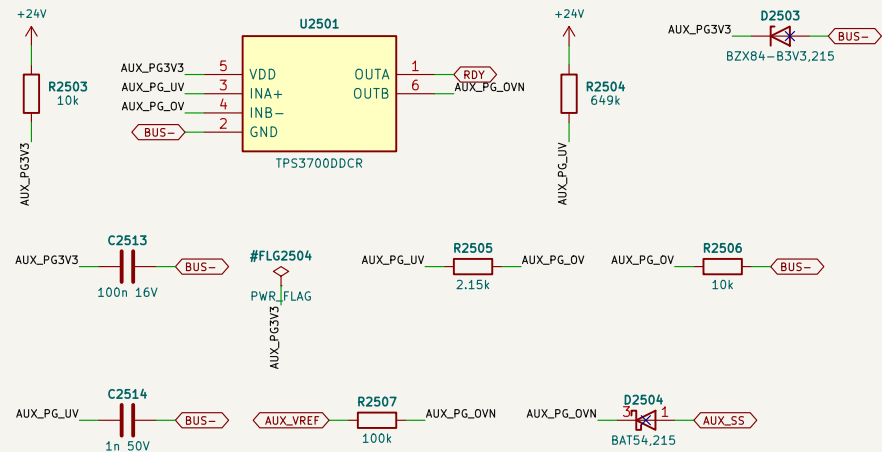
D2PAK tab = cathode on the output. Snubber 150R + 68p.

SELV 24 V RECTIFIER (REINFORCED SIDE)



Cross-regulated from the live winding.

STATUS AND OUTPUT OV STOP (TPS3700 ON A 3.3 V ZENER RAIL)



TPS3700 window on the live 24 V: UV falls 21.02–21.81 V (status), OV rises 26.13–26.76 V (SS pulled low)